Week 4 · Waypoint Following and LOS Guidance

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-09-05

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	**Control Engineering** (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	**Control System Design** (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	**Advanced Control Engineering** (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	**Sensor Signal Processing and Fusion** (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	**Capstone Design** (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

- **Course**: USV Guidance, Navigation and Control (Graduate)
- **Department**: Autonomous Vehicle System Engineering, Chungnam National University
- **This week**: where the heading command comes from; the two errors that define path following, both from one rotation; and three guidance laws — LOS, ILOS, ALOS — each derived, each with its stability argument, all four vessels racing side by side on one model

★ Prerequisites from the previous week

- Week 3 must run. `W03_0_setup` followed by `W03_C_proportional_only` should report $\zeta = 0.9000$ and a heading that settles at 60°.
- This week reuses that heading autopilot **unchanged and unretuned**. If Week 3 does not run, nothing in this week will.
- Week 3 §3-4 measured the crab angle and predicted that a path-following law would pay for it. §4-7 is that bill.

Learning Outcomes

After this week the learner should be able to:

1. Derive the along-track and cross-track errors as the two components of a single rotation, and state which of the two the guidance law regulates and which the switching logic uses.
2. Derive the LOS law from the geometry of the aim point, and state its three limits and the bound they place on the commanded turn.
3. Predict the steady cross-track error that an ocean current leaves under plain LOS, and confirm the prediction against measurement to better than 1 %.
4. Derive the ILOS integral law, explain **why its denominator has the form it has**, and identify the anti-windup that is built into the law rather than bolted onto it.
5. Derive the ALOS adaptation law from a Lyapunov function, showing that the update is **forced** by the derivation rather than chosen, and state the assumption under which the argument holds and what is lost when it fails.
6. Select Δ, R_{switch}, κ and γ from measured sweeps rather than from rules of thumb, and give the physical unit of each.
7. Explain why a guidance block that carries memory must be given a discrete sample time, and what breaks when it is not.

Prerequisites and Setup

From	What is needed here
Week 1 §1-3	$\boldsymbol{\eta} = [N\ E\ \psi]^\top, \boldsymbol{\nu} = [u\ v\ r]^\top$, and the rotation $\mathbf{R}(\psi)$
Week 1 §1-4	body velocity is not the rate of change of position — the same rotation reappears in §4-3
Week 1 §1-9	the ocean current enters as a velocity, so the vessel's speed **through the water** differs from its speed over ground
Week 3 §3-4	the crab angle and the course $\chi = \psi + \beta_c$. **Weeks 1 and 3 write this angle β; this week writes β_c,** following Fossen, because §4-9 needs β free for the estimated quantity. The two symbols mean the same thing: $\operatorname{atan2}(v, u)$
Week 3 §3-5	the P–D heading autopilot, used here **unchanged** so that the only difference between the four vessels is the guidance law
Appendix A1	the column rule and the allocation $\boldsymbol{\tau} = \mathbf{Bf}$

Software	Requirement
MATLAB / Simulink	R2023b or later
MSS toolbox	found automatically by `_tools/mss_path.m` — it is not inside this vault
Model	`lectures/W04_simulink/W04_guidance.slx`, built by `W04_1_build_guidance.m`

Part 1 · Theory

4-1. Aiming at a point is not following a path

- Weeks 1 to 3 were **told** what heading to hold. A human typed 60° into a step block and the autopilot delivered it. Nothing in those three weeks said where the number should come from.
- **Guidance** is where it comes from. Given a list of waypoints and the vessel's present position, guidance produces ψ_d; the autopilot of Week 3 then does what it already does. Guidance and control are separate layers, and this separation is the reason the Week 3 autopilot can be reused untouched.
- The obvious first law is to point the bow at the next waypoint:

$$\psi_d = \operatorname{atan2}(E_{k+1} - E,\ N_{k+1} - N)$$

Symbol	Quantity	Unit / source
$N,\ E$	present position of the vessel, NED	m — from the plant
$N_{k+1},\ E_{k+1}$	the waypoint being approached	m — `WP_N` , `WP_E` in `W04_vars.m`
ψ_d	commanded heading	rad — into the Week 3 autopilot

- This law reaches every waypoint. It is nonetheless the wrong law, and the reason is worth stating precisely: **it regulates the distance to a point, and a point carries no information about the line it sits on.**

Reaching a waypoint and following the line to it are different problems

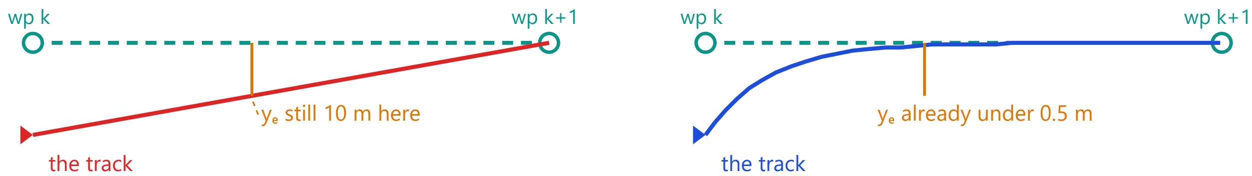
Both vessels start 18 m off the path and both arrive. Only one travels along it.

1 atan2 — aim at the waypoint

$\psi_d = \operatorname{atan2}(E_{k+1} - E,\ \ N_{k+1} - N)$

2 LOS — aim Δ ahead on the path

$\psi_d = \pi_p - \arctan(\ y_e\ /\ \Delta\ )$



The law regulates the distance to a POINT. The line the point sits on plays no part in it, so the vessel closes on the waypoint along a straight line of its own, and is on the path only at the very end — if at all.

The law regulates the distance to the LINE. The aim point moves along the path as the vessel does, so the correction fades to zero exactly when the vessel reaches the path, and the rest of the leg is run along it.

Measured on the four-leg mission of section C: atan2 holds the path to **0.55 m**, LOS to **0.06 m** — a factor of nine, and the only difference is one arctan.

Reading the figure

Element	Meaning
dashed green, both panels	the planned path — the line the vessel was asked to follow, from $\mathbf{wp}_k$ to $\mathbf{wp}_{k+1}$
left, red track	the track under `atan2` : a straight line from wherever the vessel happens to be to the waypoint
right, blue track	the track under LOS: a curve that joins the path early and then runs along it
orange ticks	the cross-track error at a comparable point of the leg — 10 m against under 0.5 m
bottom strip	the same comparison as measured on the four-leg mission of section C

- A vessel displaced from the path — by a corner, by a current, or simply by where it started — is *nearer the waypoint* than a vessel on the path. The `atan2` law therefore sees nothing wrong, and the vessel arrives from whatever direction it drifted into.
- For open water that is acceptable. For a survey line, a dredged channel, a cable route or a berth approach it is not: the mission specified a **path**, and the vehicle followed a **point**.

> **i This is a statement about the objective, not about tuning**
>
> No choice of autopilot gain makes `atan2` follow the path, because the path does not appear anywhere in the law. Section C measures the gap on the same mission with the same autopilot: on leg 2, `atan2` holds the line to 0.547 m and LOS to 0.060 m, a factor of nine. Leg 1 is excluded because the vessel starts on it, already pointing along it, so both laws read zero and the comparison is empty.

4-2. The path is a straight leg between two waypoints

- The mission is an ordered list $\mathbf{wp}_1, \ldots, \mathbf{wp}_n$ with $\mathbf{wp}_k = [N_k \ \ E_k]^\top$. Between consecutive waypoints the path is a straight line, and the **active leg** runs from $\mathbf{wp}_k$ to $\mathbf{wp}_{k+1}$.
- One number describes that leg — its direction measured from North:

$$\boxed{\pi_p = \operatorname{atan2}(E_{k+1} - E_k,\ N_{k+1} - N_k)}$$

Symbol	Quantity	Unit / source
π_p	path-tangential angle, from North, positive towards East	rad — recomputed whenever k changes
$\mathbf{wp}_k$	the leg's origin, the waypoint most recently passed	m — `WP_N(k)`, `WP_E(k)`
d_k	leg length $\lVert \mathbf{wp}_{k+1} - \mathbf{wp}_k \rVert$	m — 60, 60, 60, 84.85 m this week
n	number of waypoints	5, giving 4 legs

- The two-argument `atan2` and not `atan`: the leg may point into any of the four quadrants, and only the two-argument form separates North-East from South-West. The single-argument form would fold the third quadrant onto the first and send the vessel backwards along its own path.
- π_p is constant on a leg. Every quantity downstream of it in this week — x_e, y_e, ψ_d — changes continuously as the vessel moves, but π_p changes only at a switch.

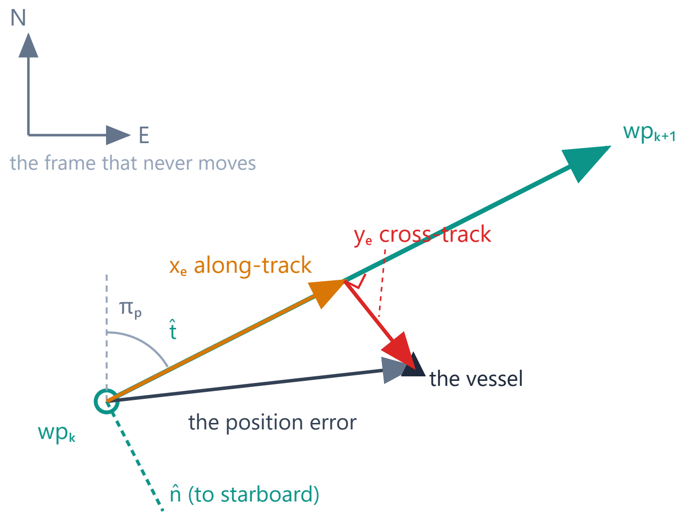
> **i Curved paths are deferred, and nothing here has to change for them**
>
> Lekkas and Fossen (2014) replace the straight leg by a Hermite spline and π_p by the spline's tangent angle at the closest point. Everything downstream of π_p in this week — §4-3 through §4-9 — is unchanged by that substitution. §4-11 gives the reference.

4-3. Along-track and cross-track error, derived

- The leg defines a frame of its own: $\hat{\mathbf{t}}$ pointing along it and $\hat{\mathbf{n}}$ pointing to starboard of it. The whole of this section is **one rotation into that frame**, and both errors fall out of it together.

Both errors come out of one rotation

Write the position error in the frame of the leg, and its two components ARE the two errors



The leg gives the frame its direction:

π_p **= atan2(** $E_{k+1} - E_k$, $N_{k+1} - N_k$ **)**

and the error, written in that frame, is

$x_e = \;\; (N - N_k)\cos\pi_p + (E - E_k)\sin\pi_p$

$y_e = -(N - N_k)\sin\pi_p + (E - E_k)\cos\pi_p$

which is $R(\pi_p)^T$ applied to the position error.

One rotation, two answers

x_e — how far ALONG the leg the vessel is. The switching test of §4-6 uses it.

y_e — how far TO THE SIDE. The guidance law drives this one to zero.

Reading the figure

Element	Meaning
grey axes, upper left	NED — the frame that does not move
green line with arrow	the active leg, from $\mathbf{wp}_k$ towards $\mathbf{wp}_{k+1}$
green dashed pair	$\hat{\mathbf{t}}$ along the leg and $\hat{\mathbf{n}}$ to starboard — the frame the leg defines
grey arrow	the position error $\mathbf{p} - \mathbf{wp}_k$, expressed in NED
orange	its component along $\hat{\mathbf{t}}$ — the along-track error x_e
red, with the right-angle mark	its component along $\hat{\mathbf{n}}$ — the cross-track error y_e
blue panel	the same statement as algebra
amber panel	which of the two errors each part of the guidance system consumes

The derivation

Written in NED, the two unit vectors of the leg frame are

$$\hat{\mathbf{t}} = \begin{bmatrix} \cos\pi_p \\ \sin\pi_p \end{bmatrix}, \qquad \hat{\mathbf{n}} = \begin{bmatrix} -\sin\pi_p \\ \cos\pi_p \end{bmatrix}$$

$\hat{\mathbf{t}}$ is a unit vector at angle π_p from North, which is the leg's direction by the definition of §4-2. $\hat{\mathbf{n}}$ is $\hat{\mathbf{t}}$ turned by $+90^\circ$, which in a North-East frame points to **starboard** of the leg. The two are orthonormal, so the pair is a basis and the decomposition below is unique.

The vessel at $\mathbf{p} = [N \;\; E]^\top$ has position error $\mathbf{p} - \mathbf{wp}_k$ relative to the leg's origin. Its components in the leg frame are its projections onto the two basis vectors:

$$x_e = \hat{\mathbf{t}}^\top(\mathbf{p} - \mathbf{wp}_k), \qquad y_e = \hat{\mathbf{n}}^\top(\mathbf{p} - \mathbf{wp}_k)$$

Stacking the two projections turns them into one matrix product, and that matrix is the transpose of the planar rotation of Week 1 §1-4:

$$\boxed{\begin{bmatrix} x_e \\ y_e \end{bmatrix} = \begin{bmatrix} \cos \pi_p & \sin \pi_p \\ -\sin \pi_p & \cos \pi_p \end{bmatrix} \begin{bmatrix} N - N_k \\ E - E_k \end{bmatrix} = \mathbf{R}(\pi_p)^\top \begin{bmatrix} N - N_k \\ E - E_k \end{bmatrix}}$$

Symbol	Quantity	Unit
x_e	along-track error — how far **along** the leg the vessel has come	m
y_e	cross-track error — how far **to the side** of the leg it is	m
$\mathbf{R}(\pi_p)$	the planar rotation from the leg frame to NED	—
$\mathbf{R}(\pi_p)^\top$	its inverse, NED to leg frame, because $\mathbf{R}$ is orthogonal	—

- Written out, the two rows are exactly the expressions printed in the figure and coded in `_tools/crosstrack_err.m`:

$$x_e = \quad (N - N_k) \cos \pi_p + (E - E_k) \sin \pi_p$$

$$y_e = -(N - N_k) \sin \pi_p + (E - E_k) \cos \pi_p$$

Verification of the derived pair

Check	Result
Dimension	both rows are metres times a dimensionless cosine — metres, as required
Limit $\mathbf{p} = \mathbf{wp}_k$	$x_e = y_e = 0$ — at the leg's origin both errors vanish
Limit $\mathbf{p} = \mathbf{wp}_{k+1}$	$x_e = d_k$, $y_e = 0$ — at the far end the vessel has run the whole leg and is on it
Sign	a point displaced by $+\varepsilon \hat{\mathbf{n}}$ gives $y_e = +\varepsilon$ — positive is to starboard
Orthogonality	$x_e^2 + y_e^2 = \lVert \mathbf{p} - \mathbf{wp}_k \rVert^2$, since $\mathbf{R}^\top$ preserves length
Source	`_tools/verify_guidance.m` compares against MSS `crosstrackWpt.m` at eight test points; largest disagreement 0, exactly

What each error is for

	consumed by	because
x_e	the **switching** logic of §4-6	it says how much of the leg has been run, so $d_k - x_e$ is how much is left
y_e	the **guidance law** of §4-4 and every law after it	it is the quantity that "follow the path" means to drive to zero

- **The sign matters and is easy to get backwards.** $y_e > 0$ places the vessel to **starboard** of the leg, because $\{n\}$ is North-East-Down and a positive rotation carries North towards East. A positive y_e must therefore be answered by turning to **port**, which is why every law in this week *subtracts* its correction from π_p. Getting this backwards produces a law that drives the vessel away from the path at a rate proportional to how far off it already is.

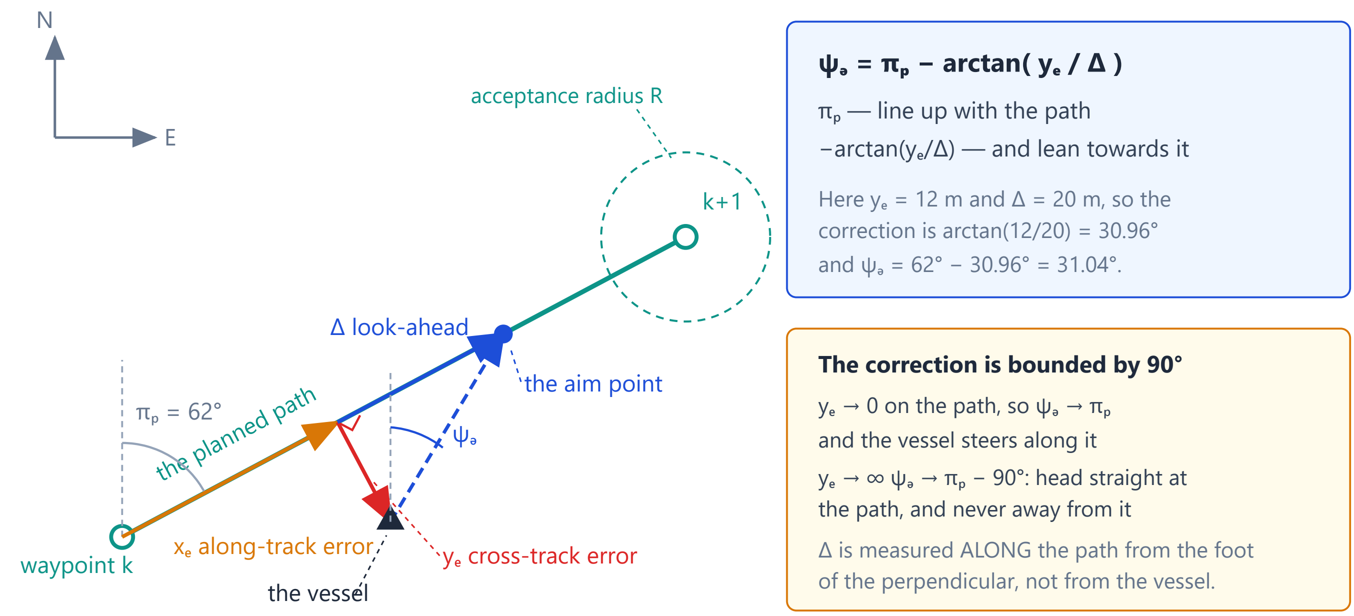
> ⚠ **Two forms are in circulation and only one of them is safe**
>
> MSS `crosstrack.m` obtains the same quantity by solving a 3×3 linear system whose coefficients contain $\tan \pi_p$. That is singular at $\pi_p = \pm 90^\circ$ — a due-East or due-West leg, which is half the legs of any lawnmower survey pattern. The rotation form above contains no tangent and has no such point. MSS `crosstrackWpt.m` uses the rotation form, and this course follows it.

4-4. The LOS law, derived

- The idea is one sentence: **do not aim at the waypoint; aim at a point on the path a fixed distance ahead of the nearest point of the path.**
- The name is literal. The vessel steers along its line of sight to a point that is always on the path and always moving forward as the vessel moves.

The line-of-sight law, and where every symbol in it lives

Aim at a point on the path, Δ ahead of the nearest one — not at the waypoint



Reading the figure

Element	Meaning
grey axes, upper left	NED, so that every angle in the figure is measured from North
green line, hollow circles	the active leg, from waypoint k to waypoint $k+1$
green dashed circle	the acceptance radius R around $\mathbf{wp}_{k+1}$, used in §4-6
grey arc, $\pi_p = 62^\circ$	the leg's direction from North
orange	x_e, from $\mathbf{wp}_k$ to the foot of the perpendicular
red, with right-angle mark	y_e, from the foot out to the vessel — the perpendicular distance to the path
blue solid	Δ, laid off **along the path** from the foot of the perpendicular
blue filled circle	the aim point, at the far end of Δ
blue dashed with arrow	the line of sight from the vessel to the aim point; its direction from North is ψ_d
blue arc at the vessel	ψ_d itself, showing that it is measured from North and not from the leg
blue panel	the law, and the arithmetic of this particular figure
amber panel	the limits, worked below

The derivation

Let F be the foot of the perpendicular from the vessel to the leg, and let the **aim point** A lie a further distance Δ along the leg. Work in the leg frame of §4-3, where the vessel sits at $(x_e,\ y_e)$ and, by construction, F sits at $(x_e,\ 0)$ and A at $(x_e + \Delta,\ 0)$.

The vector from the vessel to the aim point, **in the leg frame**, is therefore

$$A - P = \begin{bmatrix} (x_e + \Delta) - x_e \\ 0 - y_e \end{bmatrix} = \begin{bmatrix} \Delta \\ -y_e \end{bmatrix}$$

Its direction, measured from the leg's own $\hat{\mathbf{t}}$ axis, is

$$\operatorname{atan2}(-y_e,\ \Delta) = -\arctan\left(\frac{y_e}{\Delta}\right)$$

where the two-argument form collapses to the one-argument form because $\Delta > 0$ always, so the vector never leaves the right half-plane of the leg frame. The leg frame is itself rotated by π_p from North. Adding the two angles converts the direction into NED and gives the law:

$$\boxed{\psi_d = \pi_p - \arctan\left(\frac{y_e}{\Delta}\right)}$$

Symbol	Quantity	Unit	Value and origin
π_p	path-tangential angle	rad	§4-2, from the two waypoints
y_e	cross-track error	m	§4-3, from the rotation
Δ	look-ahead distance, measured **along the path**	m	8 m — chosen in §4-5, justified by the sweep of section E
y_e/Δ	argument of the arctan	—	dimensionless, as the argument of a trigonometric function must be
ψ_d	commanded heading	rad	into the Week 3 autopilot

- Read the law as two terms with two jobs. π_p says *line up with the path*. The arctan says *and lean towards it, by an amount that grows with how far off it the vessel is.*

- The figure's own arithmetic is the smallest possible worked example: $y_e = 12$ m and $\Delta = 20$ m give a correction of $\arctan(12/20) = 30.96^\circ$, so $\psi_d = 62^\circ - 30.96^\circ = 31.04^\circ$. The vessel is to starboard, so it is commanded to port, and by less than 90°.

The limits, and why they are the right ones

Limit	Result	Interpretation
$y_e \to 0$	$\psi_d \to \pi_p$	on the path, steer along it; the correction vanishes **smoothly**, so there is no chatter and no limit cycle at convergence
$y_e \to +\infty$	$\psi_d \to \pi_p - 90^\circ$	far to starboard: head **straight at** the path, perpendicular to it
$y_e \to -\infty$	$\psi_d \to \pi_p + 90^\circ$	far to port: the mirror image
$\Delta \to 0^+$	$\psi_d \to \pi_p \mp 90^\circ$ for every $y_e \neq 0$	the law degenerates to bang-bang — full correction at any error, however small
$\Delta \to \infty$	$\psi_d \to \pi_p$ for every finite y_e	the law stops correcting at all and becomes a heading hold

- The two infinite limits are the reason the law is safe to use from any initial condition: because $|\arctan(\cdot)| < 90^\circ$ strictly, **the commanded heading is never more than 90° off the path direction**. However far off the path the vessel starts, it is never commanded to turn away from it. Comparable laws built on $\psi_d = \pi_p - K y_e$ have no such bound and can command a full reversal.
- The two degenerate limits are the two ends of §4-5's trade, and section E measures both of them.

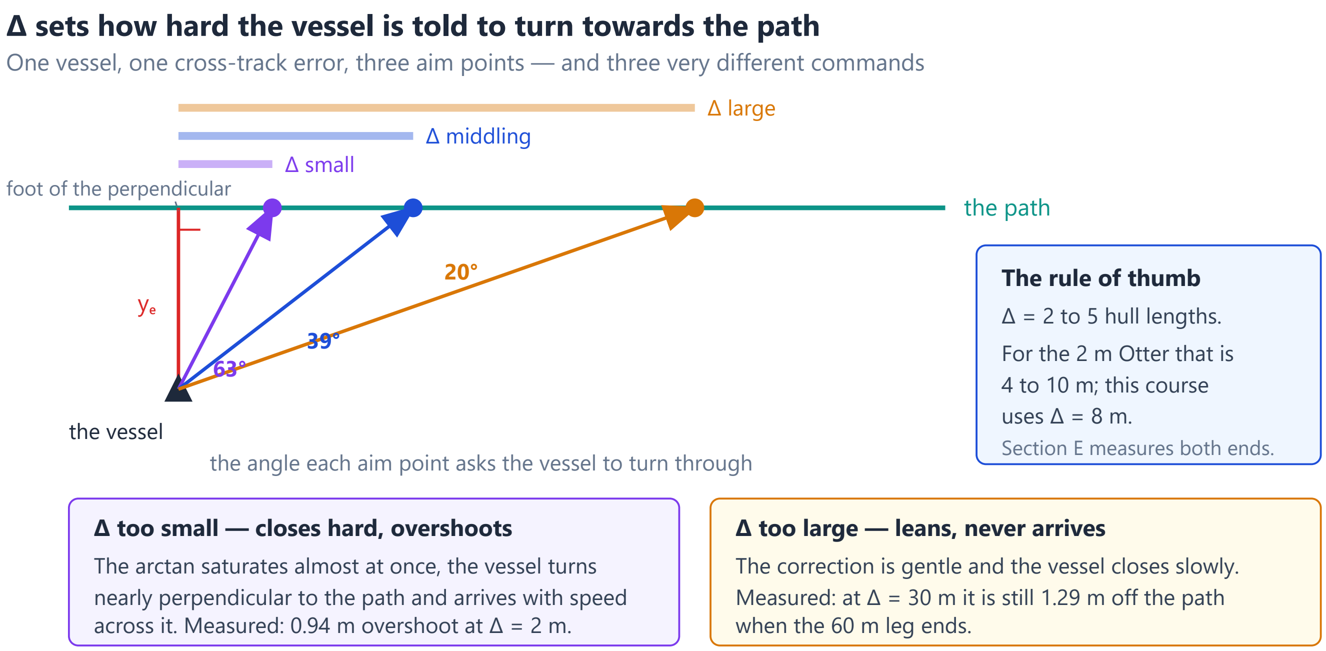
> **i Δ is measured along the path, not from the vessel**
>
> The aim point sits Δ ahead of the **foot of the perpendicular**, not at distance Δ from the boat. The distance from the boat to the aim point is $\sqrt{\Delta^2 + y_e^2}$, which is larger and which depends on y_e. Placing the aim point at a fixed range from the vessel instead makes the correction depend on y_e twice over and destroys the clean limits above.

Verification of the LOS law

Check	How, and result
Dimension	y_e/Δ is m/m; $\arctan$ of it is rad; π_p is rad — the sum is an angle
Limits	the five rows of the table above, all checked numerically in `_tools/verify_guidance.m`
Sign	$y_e = +1$ m, $\Delta = 8$ m, $\pi_p = 0$ gives $\psi_d = -7.125^\circ$ — starboard error, port command
Numeric	against MSS `LOSchi.m` on eight configurations; largest disagreement 0
Source	Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., §12.3.1

4-5. What the look-ahead distance does

- Δ is the only free number in the LOS law, and it fixes a **trade between arriving quickly and arriving smoothly**.



Reading the figure

Element	Meaning
three coloured bars, top	three values of Δ, drawn to scale against each other
green line	the path; the vessel is one and the same in all three cases
red	the cross-track error, identical in all three cases
three filled circles	the aim point each Δ produces, all on the path
three rays with arrows	the commanded heading each aim point produces
63°, 39°, 20°	how far the vessel is told to turn — **for one and the same cross-track error**
blue panel	the rule of thumb and this course's value
violet and amber panels	the two failure modes, with the numbers section E measures

- The three angles are the whole content of the figure: **the error has not changed, only Δ has, and the command has changed by a factor of three.**

Δ	behaviour of the correction	how it fails
small	saturates almost immediately, approaching 90°	closes hard and **overshoots** — measured 0.939 m of overshoot at $\Delta = 2$ m
middling	proportional over the range of errors that occur	the intended regime
large	gentle at every y_e	**never arrives** — at $\Delta = 30$ m the vessel has not reached the first leg before it ends

- Measured in section E on the same mission, autopilot and current:

Δ [m]	Δ/L	settling distance [m]	overshoot [m]	settled $\lvert y_e \rvert$ [m]
2	1.0	12.82	0.939	0.0134
4	2.0	20.06	0.630	0.0559
8	4.0	35.82	0.473	0.2321
16	8.0	never settles on leg 1	0.415	0.2796
30	15.0	never settles on leg 1	-0.679	1.2871

- The vessel starts 8 m off the path in every run, and "settling distance" is how far North it had travelled when $\lvert y_e \rvert$ last left the 0.4 m band. The two largest look-aheads never enter that band before the 60 m leg ends, so **the table reports no settling distance for them** rather than quoting the leg length as though it were one.
- Settling distance grows monotonically with Δ and overshoot falls monotonically with it: there is no value that is best at both, which is what makes this a trade rather than a tuning problem with an answer.
- The last column is the third failure. At $\Delta = 30$ m the vessel is still 1.29 m off the path at the end of the leg — **the law is no longer following the path, only leaning towards it.** A negative overshoot in that row means the same thing: the vessel never crossed the path at all.
- The common rule of thumb is Δ between 2 and 5 hull lengths. The Otter is 2 m long, so that range is 4 to 10 m. This course uses $\Delta = 8$ m — **and section E is the justification, with the rule of thumb only as a sanity check.**

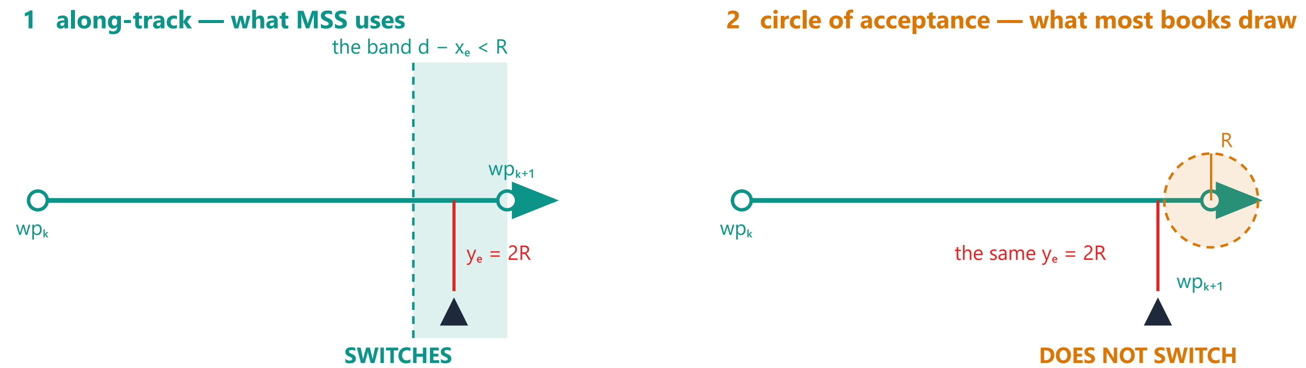
> ✎ **The third cost of a large Δ appears in §4-7**
>
> Besides slow convergence, a large Δ multiplies the steady offset that a current leaves. That is a stronger argument against large Δ than slow settling is, and it does not become visible until the current is switched on.

4-6. Deciding that a waypoint has been passed

- A leg has to end. Something must decide when, and two criteria are in common use that are **not the same criterion**.

Two ways to decide the waypoint is behind you, and they disagree

On the path they give the same answer. Off it, one switches and the other waits.



The test looks only at how much of the leg is left. Being far off the path cannot defeat it, so a vessel blown wide still moves on to the next leg.

MSS: if ($d - x_e < R\_switch$) k = k + 1

The test is a true distance, so a vessel that never enters the circle waits for a waypoint it has already passed — and can wait for ever.

if ($\lVert p - wp_{k+1} \rVert < R$) k = k + 1

Reading the figure

Element	Meaning
left panel, teal band	along-track: the band of positions where the **remaining** along-track distance $d_k - x_e$ is below R
right panel, amber circle	circle of acceptance: the disc where the **true distance** to $\mathbf{wp}_{k+1}$ is below R
black hull, both panels	the same vessel, at the same place relative to the leg, with $y_e = 2R$
verdict lines	the along-track test switches; the circle test does not
grey code lines	the two tests as they are actually written

$$\text{along-track:} \quad d_k - x_e < R \qquad\qquad \text{circle of acceptance:} \quad \|\mathbf{p} - \mathbf{wp}_{k+1}\| < R$$

Symbol	Quantity	Unit / value
d_k	length of the active leg	m — 60 m for legs 1–3 this week
x_e	along-track error of §4-3	m
$d_k - x_e$	along-track distance **remaining** on the leg	m
R	switching radius	m — `R_switch`, 5 m this week

Where the two criteria part company

- **On the path they agree**, because $y_e = 0$ makes $\|\mathbf{p} - \mathbf{wp}_{k+1}\| = d_k - x_e$ exactly. Off it they do not, because

$$\|\mathbf{p} - \mathbf{wp}_{k+1}\| = \sqrt{(d_k - x_e)^2 + y_e^2} \;\geq\; d_k - x_e$$

- The true distance is never smaller than the along-track remainder, so **the circle test is always the stricter of the two**, and it is stricter by an amount that grows with y_e. The figure's vessel has $d_k - x_e < R$ but $y_e = 2R$, so its true distance is at least $2R$ and the circle test fails.
- Section F puts numbers on the figure. A vessel 4.50 m short of waypoint 2 along the leg and 10.0 m to the side of it has a true distance of 10.97 m: the along-track test passes at $4.50 < 5$, and the circle test fails by more than a factor of two.
- The along-track test is the safer one for exactly this reason: it cannot be defeated by being far from the path. A vessel blown wide of a waypoint **never enters that waypoint's circle**, and a mission that waits for it waits for ever.
- This course uses the along-track test by default (`sw_mode = 1`), and section F runs both so that the difference can be seen rather than argued.

The upper bound on R

- R has a hard limit that has nothing to do with tuning:

$$R < \min_k d_k$$

- If R exceeded the shortest leg, the switching test for that leg would already be satisfied at the moment the leg became active — before the vessel had travelled any of it. The waypoint would be skipped without being approached, and with several short legs in a row the vehicle could skip an arbitrary number of them.
- `_tools/wp_switch.m` raises an error on it rather than silently misbehaving, as MSS does. This week's shortest leg is 60 m and $R = 5$ m, which is comfortable.

What R costs

Measured in section F on the same mission:

R [m]	corner cut [m]	switch times [s]
2	0.44	76, 154, 232
5	2.26	72, 147, 222
15	7.02	59, 128, 196
25	9.62	46, 111, 176

- Corner cut grows with R and switch times fall: a large R turns early and rounds the corner, a small R runs the leg to its end and turns sharply. The choice is a statement about whether the mission cares more about corner accuracy or about turn rate.

> ⚠ **A documentation slip in MSS, worth meeting once**
>
> The help text of `ILOSpsi.m` says that the next waypoint is taken when "the along-track distance x_e is less than R_switch". The code tests `d - x_e < R_switch`, which is the distance **remaining**, not the distance travelled. The code is the sensible one and the sentence is wrong. This is precisely why the standing rule of this course is to check an equation against the **source**, not against the description of the source.

The end of the mission

- On the last leg there is no $\mathbf{wp}_{k+2}$ to switch to, and the two implementations differ.
- MSS holds the final waypoint as **both** ends of the leg. Then $E_{k+1} - E_k = N_{k+1} - N_k = 0$, so $\pi_p = \operatorname{atan2}(0, 0) = 0$ by the IEEE convention, and the vessel is silently commanded due North for ever.
- This course stops the leg index at $n - 1$ and keeps the last leg active, so the vessel continues along its extension. The behaviour is at least predictable.
- **Neither of these is a mission-termination policy.** A real vehicle needs one — hold station at the last waypoint, loiter around it, or stop the propellers — and the choice belongs to the mission, not to the guidance law. Week 7 puts a Stateflow chart in charge of it.

4-7. Heading is not course — the debt from Week 3

- Week 3 §3-4 measured the crab angle and warned that a path-following law would eventually pay for it. This section is that bill, and it is the reason the two remaining laws of this week exist.
- LOS commands a **heading**, and the autopilot delivers that heading faithfully. But the vessel travels along its **course**, which differs from its heading by the crab angle:

$$\chi = \psi + \beta_c, \qquad \beta_c = \operatorname{atan2}(v,\, u)$$

Symbol	Quantity	Unit
ψ	heading — where the bow points	rad
χ	course — the direction the vessel actually moves over ground	rad
β_c	crab angle, over ground, from the Week 3 definition	rad
U	speed over ground, $\sqrt{u^2 + v^2}$	m/s

The cross-track error dynamics

- Before the offset can be derived, the rate of change of y_e is needed, and it is used again in §4-8 and §4-9.

- The vessel's velocity over ground has magnitude U and direction χ, both measured in NED. Its component along $\hat{\mathbf{n}}$ — which is $\hat{\mathbf{t}}$ turned by 90°, so the angle between the velocity and $\hat{\mathbf{n}}$ is $\chi - \pi_p - 90^\circ$ — gives the rate at which the cross-track error grows:

$$\boxed{\dot{y}_e = U\sin(\chi - \pi_p)}$$

Check	Result
Dimension	m/s on both sides
Limit $\chi = \pi_p$	$\dot{y}_e = 0$ — travelling along the path, the offset is frozen at whatever it is
Limit $\chi = \pi_p + 90^\circ$	$\dot{y}_e = U$ — travelling straight to starboard, the offset grows at full speed
Sign	$\chi > \pi_p$ turns the velocity towards starboard and increases y_e, consistent with §4-3

- Note what this equation contains: U and χ, both **over ground**. The current has already been absorbed into them, which is why no current term appears explicitly.

The steady offset, derived

In steady state on a straight leg the cross-track error has stopped changing, which by the equation above requires the **course** to lie along the path:

$$\dot{y}_e = 0 \quad\Longrightarrow\quad \sin(\chi - \pi_p) = 0 \quad\Longrightarrow\quad \chi = \pi_p$$

The autopilot has by then delivered its command, $\psi = \psi_d$, and ψ_d is the LOS law of §4-4. Substituting both:

$$\pi_p \;=\; \chi \;=\; \psi + \beta_c \;=\; \psi_d + \beta_c \;=\; \pi_p - \arctan\left(\frac{y_e}{\Delta}\right) + \beta_c$$

The π_p cancels from both sides — which is the point, because it means the result does not depend on which leg the vessel is on:

$$\arctan\left(\frac{y_e}{\Delta}\right) = \beta_c$$

and therefore

$$\boxed{y_e^{ss} = \Delta\,\tan\beta_c}$$

Reading	
What it says	the vessel settles **parallel to the path and beside it**, offset by an amount proportional to Δ
Why it is not a tuning failure	the loop is doing exactly what it was asked. It is holding ψ_d to the last decimal, and ψ_d was computed by a law in which β_c never appears
Why more autopilot gain does not help	the heading error is already zero. There is nothing left for the autopilot to act on
The third cost of a large Δ	the offset is *linear* in Δ: doubling the look-ahead doubles the steady error. §4-5's trade has a third term after all

Verification against measurement

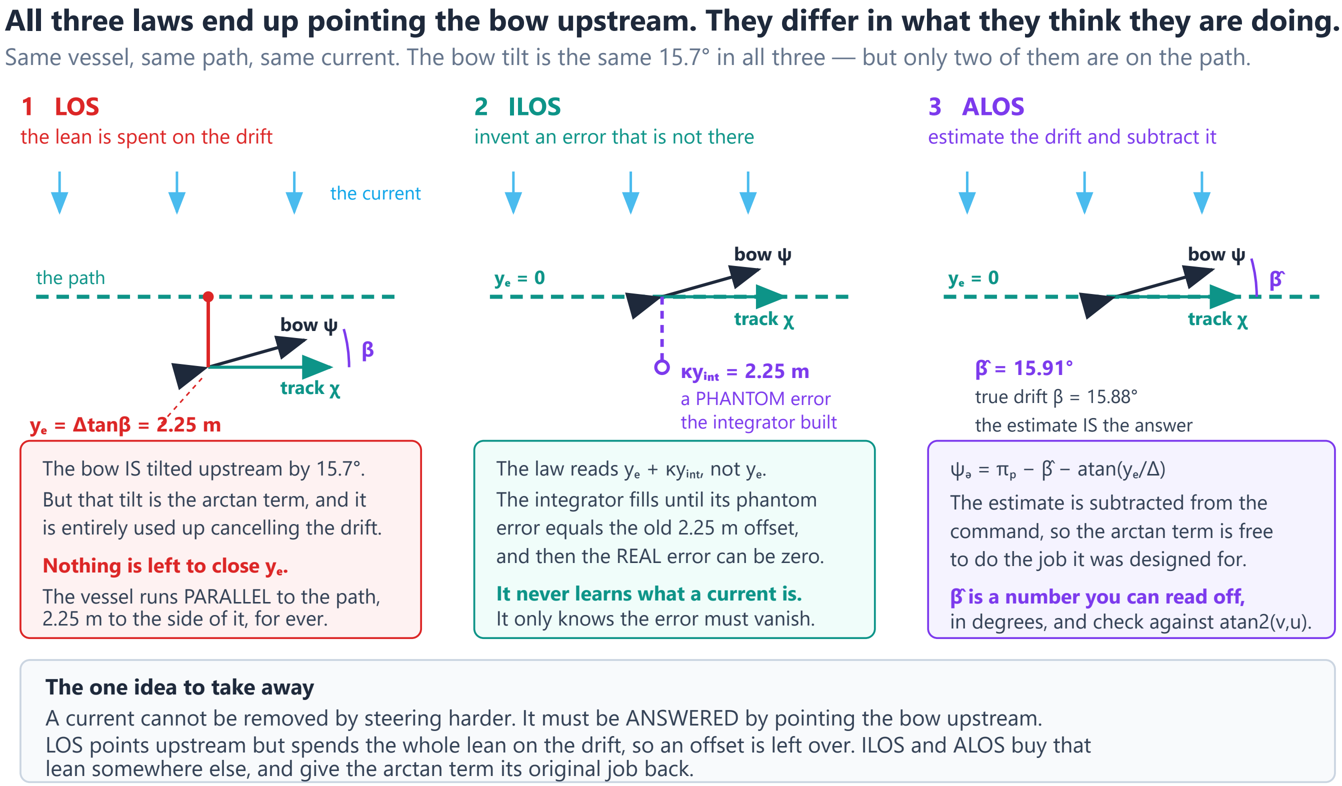
Section G sweeps the current speed with everything else held fixed and compares the measured settled offset against the prediction:

V_c [m/s]	β_c [deg]	measured y_e [m]	$\Delta \tan \beta_c$ [m]	difference [m]
0.0	-0.10	-0.014	-0.013	0.001
0.1	5.10	0.713	0.714	0.001
0.2	10.37	1.463	1.464	0.001
0.3	15.74	2.253	2.254	0.001
0.4	21.26	3.111	3.112	0.001
0.5	27.38	4.145	4.143	0.002

- The largest disagreement across the whole sweep is 0.002 m on an offset of 4.1 m — better than one part in two thousand. The derivation is not an approximation; it is what the loop does.
- The remainder of this week is two different ways of removing this offset. §4-8 integrates it away without ever learning what caused it. §4-9 estimates the cause and cancels it directly.

Three answers, in one picture

- Before either derivation, it is worth seeing what the two remaining laws are *for*. The figure below is the whole of §4-8 and §4-9 with no algebra in it.



Reading the figure

Element	Meaning
blue arrows, all three panels	the current, identical in each
dashed green line	the path
black hull and arrow	where the **bow** points — the heading ψ
solid green arrow	where the vessel actually **travels** — the course χ
violet arc	the angle between them, the crab angle
panel 1, red line	the steady offset $\Delta \tan \beta = 2.25$ m that plain LOS is left holding
panel 2, violet dashed line	the **phantom** cross-track error the ILOS integrator has built. There is no vessel there
panel 3, violet arc	the estimate $\hat{\beta}$, subtracted from the command

What the figure says

- **Meaning.** One vessel, one path, one current, three guidance laws. The question each panel answers is: *where does the bow end up pointing, and is the vessel on the path?*
- **The one thing that is the same in all three.** The bow is tilted upstream by 15.7° in every panel. **That tilt is not optional** — it is the only way to travel along a path while water pushes sideways, and any law that works must produce it. Comparing the three laws is comparing *how they pay for the same tilt*.
- **Panel 1, and why LOS is stuck.** LOS has exactly one way to tilt the bow: the $\arctan(y_e/\Delta)$ term. Using it to cancel the drift means it is no longer available to close the gap, so the vessel ends up parallel to the path and 2.25 m beside it. **The law is not short of authority; it is short of terms.**
- **Panel 2, the trick ILOS plays.** The integrator adds a second quantity inside the same arctan. Once that quantity has grown to $\kappa y_{int} = 2.25$ m, the law is being told there is a 2.25 m error even though the vessel is exactly on the path — so it keeps the bow tilted while the real error sits at zero. **The integrator is a lie the law tells itself, and the lie is exactly the size of the truth it replaced.**
- **Panel 3, what ALOS does instead.** ALOS adds a term *outside* the arctan: it subtracts an estimate of the drift angle directly from the command. The bow tilts by $\hat{\beta}$, and the arctan term is handed back its original job of closing y_e. **Nothing is invented; the disturbance is measured and removed.**
- **What separates the two.** Both end with the same picture — on the path, bow upstream — and the state each carries is what differs. ILOS carries a number with no meaning (7.52, in seconds); ALOS carries the crab angle itself (15.91° against a true 15.88°). **Only one of them can be checked against a measurement.**

> **✎ If only one sentence from this week is remembered**
>
> A current is not resisted, it is **answered**. The vessel must point upstream, and the three laws differ only in where they find the authority to do it.

4-8. ILOS — integral line of sight, derived

> **★ Sections 4-8 and 4-9 are worked to a different standard**
>
> These are the two advanced laws of the course, and for them **every equation and every parameter is derived rather than quoted**. Each section gives the error dynamics first, then the reason for the particular Lyapunov function and the particular weight in it, then the derivative expanded line by line with nothing skipped, then the point at which the update law is *forced* rather than chosen, then a table of every parameter with its unit, and finally the assumptions and what is lost when they fail.

4-8-1. The problem, restated as an equation

- §4-7 established that plain LOS settles at $y_e^{ss} = \Delta \tan \beta_c$, and that this happens with the heading error already at zero. The offset is not a tracking failure; it is what the law asks for.
- The classical fix for a steady offset is an integrator, and the classical failure of that fix is windup. The law of this section does both at once: it adds the integrator **inside the arctan**, which turns out to make the anti-windup part of the law rather than something bolted on afterwards.

4-8-2. The law

Børhaug, Pavlov and Pettersen (2008) replace the LOS argument by a proportional-plus-integral combination:

$$\boxed{\psi_d = \pi_p - \arctan\big(K_p\, y_e + K_i\, y_{int}\big), \qquad K_p = \frac{1}{\Delta}, \quad K_i = \kappa K_p}$$

$$\boxed{\dot{y}_{int} = \frac{\Delta\, y_e}{\Delta^2 + \big(y_e + \kappa\, y_{int}\big)^2}}$$

- **Where $K_p = 1/\Delta$ comes from.** It is not a new gain. The plain LOS law of §4-4 is $\psi_d = \pi_p - \arctan(y_e/\Delta)$, and writing y_e/Δ as $K_p y_e$ merely names the coefficient. The proportional gain of *every* LOS law in this week is $1/\Delta$, in units of m^{-1}, and §4-5's trade is therefore a statement about a proportional gain.
- **Where $K_i = \kappa K_p$ comes from.** Writing the integral gain as a multiple of the proportional gain makes κ the single tuning number, and it makes the whole argument of the arctan collapse to

$$K_p y_e + K_i y_{int} = \frac{y_e + \kappa\, y_{int}}{\Delta}$$

which is the same y_e/Δ as before with y_e replaced by $y_e + \kappa y_{int}$. **The integral term acts as a fictitious extra cross-track error**, and that is the cleanest way to hold the law in mind. Write

$$a \;\triangleq\; y_e + \kappa\, y_{int}, \qquad D \;\triangleq\; \sqrt{\Delta^2 + a^2}$$

so that the law is $\psi_d = \pi_p - \arctan(a/\Delta)$ and the update is $\dot{y}_{int} = \Delta y_e / D^2$.

4-8-3. Every parameter, with its unit

Symbol	Quantity	Unit	Where it comes from
Δ	look-ahead distance	m	§4-5; 8 m, from the sweep of section E
K_p	proportional gain, $1/\Delta$	m^{-1}	**not free** — fixed by Δ
κ	integral gain constant	**m/s**	§4-8-7; 0.3 m/s, from the sweep of section H
K_i	integral gain, κ/Δ	s^{-1}	**not free** — fixed by Δ and κ
y_{int}	integral state	**s**	a state of the guidance block
a	$y_e + \kappa y_{int}$	m	the effective cross-track error
D	$\sqrt{\Delta^2 + a^2}$	m	distance from vessel to aim point, with a in place of y_e

⚠ **The units of κ and y_{int} are not what most readers assume**

$\dot{y}_{int} = \Delta y_e/(\Delta^2 + a^2)$ has metres times metres over metres squared, so $\dot{y}_{int}$ **is dimensionless** and y_{int} carries the unit of **seconds**. For κy_{int} to be a length, κ must therefore be a **speed**, in m/s.
The course-angle sibling `ILOSchi.m` normalises differently — $\dot{y}_{int} = U y_e/\sqrt{\Delta^2 + a^2}$, which is $(\mathrm{m/s}) \cdot \mathrm{m/m} = \mathrm{m/s}$ — so there y_{int} is a **length** and κ is **dimensionless**. The same symbol κ means two different physical quantities in two files of the same toolbox. A value of κ carried from one to the other is not merely mistuned, it is dimensionally wrong. §4-8-6 explains why the two normalisations exist.

4-8-4. Why the denominator has that form — the anti-windup is in the law

- Compare the update with a plain integrator, $\dot{y}_{int} = y_e$. The ILOS update is that plain integrator multiplied by a scaling factor:

$$\dot{y}_{int} = \underbrace{y_e}_{\text{plain integral}} \times \underbrace{\frac{\Delta}{\Delta^2 + a^2}}_{\text{the scaling}}$$

- The scaling depends on a, and a is **exactly the argument of the arctan in the law**. That is the whole design: the integrator is throttled by how saturated the guidance law is.

regime	$\lvert a\rvert$ against Δ	scaling	behaviour
linear	$\lvert a\rvert \ll \Delta$	$\to 1/\Delta$	a plain integrator with gain $1/\Delta$
knee	$\lvert a\rvert = \Delta$	$1/(2\Delta)$	half rate — the arctan is at 45°
saturated	$\lvert a\rvert \gg \Delta$	$\to \Delta/a^2$	the integrator all but stops

Computed for $\Delta = 8$ m:

$\lvert a\rvert/\Delta$	scaling	as a fraction of $1/\Delta$
0.01	0.12499	0.9999
0.10	0.12376	0.9901
1.00	0.06250	0.5000
5.00	0.00481	0.0385
20.00	0.00031	0.0025

- **This is the textbook definition of anti-windup**, and it arrives without a saturation block, without a clamp, and without a back-calculation gain. Week 2 §F built anti-windup by hand for the surge integrator, with a limiter and a feedback path; here the same effect is a property of the equation.
- The rate is also bounded. With y_{int} following y_e in sign, as it does in operation,

$$\max_{y_e} \frac{\Delta\, y_e}{\Delta^2 + y_e^2} = \frac{1}{2} \quad \text{at} \quad y_e = \Delta$$

verified numerically as 0.500000. The integral state can never run away faster than half a second per second, whatever the cross-track error.

4-8-5. The equilibrium — where the offset goes

- Set both derivatives to zero. The integrator condition comes first and it is decisive:

$$\dot{y}_{int} = \frac{\Delta\, y_e}{D^2} = 0 \quad \Longrightarrow \quad y_e = 0$$

because $\Delta > 0$ and $D^2 \geq \Delta^2 > 0$, so the fraction vanishes only through its numerator. **The integrator has no equilibrium other than zero cross-track error.** This one line is the entire reason for adding it, and it holds regardless of κ, Δ, the current, or the vessel.

- Now the position condition. From §4-7, $\dot{y}_e = 0$ requires $\chi = \pi_p$, and $\chi = \psi_d + \beta_c$ once the autopilot has converged:

$$\pi_p = \pi_p - \arctan\left(\frac{y_e + \kappa y_{int}}{\Delta}\right) + \beta_c$$

With $y_e = 0$ from the first condition,

$$\arctan\left(\frac{\kappa\, y_{int}^{eq}}{\Delta}\right) = \beta_c \quad \Longrightarrow \quad \boxed{y_{int}^{eq} = \frac{\Delta \tan \beta_c}{\kappa}}$$

- Read the last line against §4-7. The quantity $\kappa y_{int}^{eq} = \Delta \tan \beta_c$ is **exactly the offset that plain LOS was left holding**. The integrator does not remove the crab angle and does not know it exists; it accumulates until the fictitious cross-track error it contributes equals the real one that the crab angle was producing, and from then on the real one can be zero.

Verification	Result
$\Delta = 8$ m, $\kappa = 0.3$ m/s, $\beta_c = 15.74^\circ$	$y_{int}^{eq} = 7.5158$ s
κy_{int}^{eq} against $\Delta \tan \beta_c$	2.254726 m against 2.254726 m — difference exactly 0
both derivatives evaluated at the equilibrium	$\dot{y}_e = 0$, $\dot{y}_{int} = 0$, to machine zero
integrating the pair for 400 s from $y_e = 5$ m, $y_{int} = 0$	$y_e \to 1.05 \times 10^{-9}$ m, $y_{int} \to 7.5158$ s — the predicted value
measured on the full model, section G	$y_e = 0.008$ m against LOS's 2.253 m

4-8-5a. Watching the integrator fill

- The equilibrium above says *where* the integrator ends. This table says *how it gets there*, and it is the clearest way to see what the law is doing. It integrates the error dynamics of §4-8-6 directly, with no autopilot in the loop, so it is the idealised picture rather than the full model.
- Printed by `W04_G_current_and_integral.m`, at the end of its output.

t [s]	y_e [m]	y_{int} [s]	κy_{int} [m]	bow tilt [deg]
0	0.000	0.000	0.000	0.00
10	1.622	1.289	0.387	14.09
25	1.337	3.997	1.199	17.59
50	0.472	6.440	1.932	16.72
100	0.036	7.437	2.231	15.82
200	0.000	7.515	2.255	**15.74**
400	0.000	7.516	2.255	**15.74**

- Read the table left to right and then top to bottom.

What to notice	Why it happens
y_e **grows first**, to 1.62 m at $t = 10$ s	at $t = 0$ the integrator is empty, so the law is plain LOS and the current pushes the vessel off the path exactly as §4-7 says it must
κy_{int} climbs towards 2.255	the fourth column is the phantom error of the figure, filling up
bow tilt **overshoots** to 17.59° at $t = 25$ s	the integrator does not know when to stop; it overshoots and comes back, which is why §4-8-8's sweep has an interior minimum
the last column settles on 15.74°	which is β_c — **the tilt the vessel needed all along**, now supplied by the integrator instead of by a standing error
$\kappa y_{int} \to 2.255$ m	the same $\Delta \tan \beta_c$ that plain LOS carried as a *real* offset, now carried as a *phantom* one

- **The last two rows are the whole idea.** The vessel is on the path, $y_e = 0.000$, and the bow is still tilted 15.74° upstream. Plain LOS could only produce that tilt by being 2.255 m off the path. ILOS produces it from a state instead.

4-8-6. Stability — and why there are two normalisations

- To argue stability, put the closed loop into error coordinates. Let $\tilde{y} = y_{int} - y_{int}^{eq}$ be the integrator's error, and assume for the argument that the autopilot is fast enough that $\psi = \psi_d$ and that β_c is constant. Starting from §4-7's $\dot{y}_e = U \sin(\chi - \pi_p)$ with $\chi - \pi_p = \beta_c - \arctan(a/\Delta)$:

$$\dot{y}_e = U \sin\Big(\beta_c - \arctan \frac{a}{\Delta}\Big)$$

Expand the sine of a difference, and use $\cos(\arctan(a/\Delta)) = \Delta/D$ and $\sin(\arctan(a/\Delta)) = a/D$:

$$\dot{y}_e = U \left[\sin\beta_c \cdot \frac{\Delta}{D} - \cos\beta_c \cdot \frac{a}{D}\right] = \frac{U}{D}\Big[\Delta \sin\beta_c - a\cos\beta_c\Big]$$

Substitute $a = y_e + \kappa y_{int} = y_e + \kappa\tilde{y} + \Delta\tan\beta_c$, using the equilibrium value from §4-8-5:

$$\Delta\sin\beta_c - \big(y_e + \kappa\tilde{y} + \Delta\tan\beta_c\big)\cos\beta_c = \underbrace{\Delta\sin\beta_c - \Delta\tan\beta_c\cos\beta_c}_{=0} - (y_e + \kappa\tilde{y})\cos\beta_c$$

The bracketed pair cancels identically, because $\tan\beta_c \cos\beta_c = \sin\beta_c$. **The current disappears from the error dynamics**, which is the formal statement of what the integrator achieved:

$$\boxed{\dot{y}_e = -\frac{U\cos\beta_c}{D}\,(y_e + \kappa\tilde{y})}, \qquad \dot{\tilde{y}} = \frac{\Delta\, y_e}{D^2}$$

Verification	Result
the closed form against the original $U\sin(\beta_c - \arctan(a/\Delta))$	agreement to 6.7×10^{-16} over 10^4 random states

Choosing the Lyapunov function

- The state is $(y_e, \tilde{y})$ and both should go to zero, so the natural candidate is a weighted sum of squares:

$$V = \tfrac{1}{2}y_e^2 + \tfrac{c}{2}\tilde{y}^2, \qquad c > 0$$

- V is positive definite and radially unbounded for any $c > 0$. The weight c is not decoration: it is the **one free quantity**, and the entire argument turns on whether a constant c exists that removes the indefinite cross term. Differentiating along the trajectories, with nothing omitted:

$$\dot{V} = y_e\dot{y}_e + c\,\tilde{y}\,\dot{\tilde{y}}$$

$$\dot V = y_e\left[-\frac{U\cos\beta_c}{D}(y_e+\kappa\tilde y)\right] + c\,\tilde y\left[\frac{\Delta y_e}{D^2}\right]$$

$$\dot V = -\frac{U\cos\beta_c}{D}\,y_e^2 \underbrace{-\,\frac{U\kappa\cos\beta_c}{D}\,y_e\tilde y \;+\; \frac{c\,\Delta}{D^2}\,y_e\tilde y}_{\text{the cross term}}$$

- The first term is what is wanted: negative whenever $y_e \neq 0$, provided $\cos\beta_c > 0$. The cross term has no definite sign and must be made to vanish. Collecting it,

$$y_e\tilde y\left[\frac{c\Delta}{D^2} - \frac{U\kappa\cos\beta_c}{D}\right] = 0 \quad\iff\quad c = \frac{U\kappa\,D\cos\beta_c}{\Delta}$$

- **and this is where the heading form runs out of room**: D is a function of the state, so the required c is not constant and V as written is not a Lyapunov function. Substituting the constant $c = \kappa\cos\beta_c$ and measuring what is left over 10^4 random states gives a residual with rms 2.94 — it is genuinely non-zero, not a small error.

The course form closes where the heading form does not

- Repeat the calculation with the normalisation of `ILOSchi.m`, $\dot{\tilde y} = U y_e/D$. Only the second term of $\dot V$ changes:

$$\dot V = -\frac{U\cos\beta_c}{D}y_e^2 - \frac{U\kappa\cos\beta_c}{D}y_e\tilde y + c\,\frac{U}{D}y_e\tilde y$$

Now both cross terms carry the same $1/D$, and the choice

$$c = \kappa\cos\beta_c \;>\; 0 \quad \text{for } |\beta_c| < 90^\circ$$

is a genuine constant. It cancels them exactly and leaves

$$\boxed{\dot V = -\,\frac{U\cos\beta_c}{D}\,y_e^2 \;\leq\; 0}$$

Verification	Result
$\dot V$ against $-U\cos\beta_c\,y_e^2/D$, course form	agreement to 1.1×10^{-14} over 10^4 random states
$\dot V \leq 0$ over all 10^4 samples	true

- $\dot V \leq 0$ gives stability and boundedness. It is only negative *semi*definite — $\dot V = 0$ on the whole line $y_e = 0$ — so LaSalle's invariance principle supplies the rest: on that line $\dot y_e = -U\cos\beta_c\,\kappa\tilde y/D$, which is non-zero unless $\tilde y = 0$, so the largest invariant set inside $\dot V = 0$ is the single point $(0,0)$, and the equilibrium is asymptotically stable.

> **i What this does and does not settle**
>
> The clean cancellation above is for the **course** form. The heading form of `ILOSpsi.m` differs from it only by the positive state-dependent factor $\Delta/(UD)$ multiplying the integrator, so it has the **same equilibrium** and the **same sign of integration** — §4-8-5 is untouched, and the numerical integration of §4-8-5 confirms convergence. What it does not have is this one-line quadratic proof. Børhaug, Pavlov and Pettersen (2008) prove the heading case by a cascade argument instead, and their result is uniform global asymptotic stability of the (y_e, y_{int}) subsystem under a bound on κ.
> The two normalisations in MSS are therefore not an inconsistency. They are two laws for two different autopilots, and each is normalised so that *its own* stability argument closes.

4-8-7. The assumptions, and what breaks without them

Assumption	Used where	What fails if it does not hold
$\psi = \psi_d$ — the autopilot is infinitely fast	$\chi = \psi_d + \beta_c$ in §4-8-5 and §4-8-6	with real autopilot lag the true system is a cascade; V is no longer monotone during transients. §4-9-7 measures exactly this
β_c constant	the cancellation $\Delta \sin \beta_c - \Delta \tan \beta_c \cos \beta_c = 0$	a time-varying current leaves a residual proportional to $\dot{\beta}_c$; the integrator tracks it with a lag
$\lvert\beta_c\rvert < 90^\circ$	$\cos \beta_c > 0$, the sign of both the leading term and the weight c	the vessel is moving backwards relative to its heading; the law's sign inverts
$U > 0$	$\dot{V} < 0$	at rest there is no guidance at all — the law commands a heading but nothing moves along the path
$\Delta > 0$	$D \geq \Delta > 0$, and $K_p = 1/\Delta$	division by zero
straight leg, π_p constant	$\dot{y}_e = U \sin(\chi - \pi_p)$	on a curved path an extra curvature term appears; Lekkas and Fossen (2014) carry it

4-8-8. Choosing κ

- κ sets how fast the integrator fills. From §4-8-5 the state must reach $\Delta \tan \beta_c / \kappa$, and from §4-8-4 it fills at up to $1/2$ per second, so the time to converge scales roughly as $\Delta \tan \beta_c / (2\kappa) \cdot 2 = \Delta \tan \beta_c / \kappa$ — **a larger κ needs a smaller final state and therefore converges sooner**, at the cost of a larger contribution per unit of y_{int} and hence more overshoot.
- Section H measures it on the full model rather than arguing it. Mean absolute cross-track error over the final 100 s:

κ [m/s]	K_i [s^{-1}]	settled $\lvert y_e\rvert$ [m]	peak $\lvert y_e\rvert$ [m]	settling time [s]
0.02	0.0025	1.3425	3.6388	never
0.05	0.0063	0.6742	3.4835	never
0.10	0.0125	0.2033	3.2855	never
0.30	0.0375	**0.0103**	2.8057	66.9
1.00	0.1250	0.0260	2.1178	73.2

- The curve has an interior minimum at $\kappa = 0.3$, which is the signature of a genuine trade rather than a monotone gain. Below it the integral state has not finished filling by the end of the run — the "never" entries are not instability but an integrator still climbing. Above it the loop begins to ring and the settled error grows again.
- The peak column moves the other way: a larger κ pulls harder and so *reduces* the worst excursion while *worsening* the settled value. Choosing κ is therefore a choice about which of those two matters.
- **This course uses $\kappa = 0.3$ m/s because the sweep says so**, not because it is a round number.

4-9. ALOS — adaptive line of sight, derived

⚠ The source situation for this section, stated plainly

The MSS release vendored for this course is the **2021** one. It contains `ILOSpsi.m` and `ILOSchi.m` but **no** `ALOSpsi.m` — adaptive LOS entered MSS in 2023. There is therefore no reference implementation to check against, and the exact equation could not be extracted from the available Fossen PDFs in a form that could be quoted with confidence.

The derivation below was therefore carried out here from first principles, and it is verified numerically instead of by citation: the adaptation law is shown to be forced by the Lyapunov argument, and the estimate it produces is checked against the true crab angle in simulation (§4-9-6). Where this differs from Fossen (2023) in scaling, the derivation, not the citation, is what this course stands behind.

4-9-1. A different idea

- ILOS removes the offset **without ever learning what caused it**. Its integral state ends at whatever value makes the error zero, and if asked what the current was doing, it cannot say.
- ALOS asks a different question: since §4-7 showed that the entire problem is the unknown angle β, why not **estimate β and subtract it**? The estimate then has physical meaning — it is the crab angle, in degrees, available as an output.

4-9-2. The law

$$\boxed{\psi_d = \pi_p - \hat{\beta} - \arctan\left(\frac{y_e}{\Delta}\right)}$$

$$\boxed{\dot{\hat{\beta}} = \gamma\,\frac{\Delta\, y_e}{\sqrt{\Delta^2 + y_e^2}}}$$

- The guidance term is the plain LOS law of §4-4 with one extra term: the estimate of the crab angle, subtracted so that the *course* rather than the *heading* points where LOS wants it.
- The rest of this section derives the second equation. It is not a design choice — the Lyapunov argument leaves no alternative.

4-9-3. Every parameter, with its unit

Symbol	Quantity	Unit	Where it comes from
Δ	look-ahead distance	m	§4-5; 8 m
$\hat{\beta}$	estimate of the crab angle	rad	a state of the guidance block
β	the true crab angle — **the same β_c used in §4-7 and §4-8**, written without the subscript here so that $\hat{\beta}$ and $\tilde{\beta}$ stay legible	rad	unknown to the law; measured only for verification
$\tilde{\beta}$	estimation error $\beta - \hat{\beta}$	rad	appears only in the analysis
γ	adaptation gain	**rad/(m·s)**	§4-9-8; 0.005, from the sweep of section H
U	speed over ground	m/s	≈ 1.31 m/s for the Otter at this thrust

- **The unit of γ is worth checking**, because it is the one number a reader is likely to carry across from another vehicle. $\dot{\hat{\beta}}$ is rad/s. The fraction $\Delta y_e / \sqrt{\Delta^2 + y_e^2}$ has metres times metres over metres, so it is a **length**. For the product to be rad/s, γ must be $\mathrm{rad}/(\mathrm{m}\cdot\mathrm{s})$. It is not dimensionless, and a value tuned on a vehicle of a different size is not transferable without rescaling.

4-9-4. The error dynamics

- Start from §4-7, unchanged:

$$\dot{y}_e = U\sin(\chi - \pi_p), \qquad \chi = \psi + \beta$$

- Assume the autopilot has converged, $\psi = \psi_d$, and substitute the ALOS law:

$$\chi - \pi_p = \psi_d + \beta - \pi_p = \left[\pi_p - \hat{\beta} - \arctan\frac{y_e}{\Delta}\right] + \beta - \pi_p = \tilde{\beta} - \arctan\frac{y_e}{\Delta}$$

using $\tilde{\beta} = \beta - \hat{\beta}$. The path direction has cancelled, and the cross-track dynamics reduce to

$$\boxed{\dot{y}_e = U\sin\left(\tilde{\beta} - \arctan\frac{y_e}{\Delta}\right)}$$

- **Read this before going on.** If the estimate were perfect, $\tilde{\beta} = 0$ and this is the plain LOS convergence of §4-4 with no offset at all. The whole problem has been reduced to driving one scalar, $\tilde{\beta}$, to zero.
- The current is unknown but constant, so $\dot{\beta} = 0$ and therefore

$$\dot{\tilde{\beta}} = -\dot{\hat{\beta}}$$

which is the equation that lets the adaptation law enter the Lyapunov derivative at all.

4-9-5. The Lyapunov function, and why its weight is what it is

- Two quantities must go to zero and they have different units — y_e in metres, $\tilde{\beta}$ in radians. A Lyapunov function has to add them, so it must carry a weight that reconciles the units, and the weight is not free once the cancellation is demanded.

$$V = \tfrac{1}{2}y_e^2 + \frac{U}{2\gamma}\tilde{\beta}^2$$

Term	Unit	Why this weight
$\frac{1}{2}y_e^2$	m^2	the quantity actually being regulated
$\frac{U}{2\gamma}\tilde{\beta}^2$	$\dfrac{\mathrm{m/s}}{\mathrm{rad}/(\mathrm{m\,s})}\mathrm{rad}^2 = \mathrm{m}^2$	the $1/\gamma$ is what makes γ cancel out of $\dot{V}$; the U is what makes the two cross terms carry the same factor and so become cancellable at all

- V is positive definite in $(y_e, \tilde{\beta})$ and radially unbounded, for any $\gamma > 0$ and $U > 0$.
- Differentiate, treating U as constant:

$$\dot{V} = y_e\,\dot{y}_e + \frac{U}{\gamma}\,\tilde{\beta}\,\dot{\tilde{\beta}} = y_e\,U\sin\left(\tilde{\beta} - \arctan\frac{y_e}{\Delta}\right) - \frac{U}{\gamma}\,\tilde{\beta}\,\dot{\hat{\beta}}$$

- Expand the sine of a difference. Writing $b = \arctan(y_e/\Delta)$, the right triangle of §4-4 gives

$$\cos b = \frac{\Delta}{\sqrt{\Delta^2 + y_e^2}}, \qquad \sin b = \frac{y_e}{\sqrt{\Delta^2 + y_e^2}}$$

so that

$$\sin(\tilde{\beta} - b) = \sin\tilde{\beta}\cos b - \cos\tilde{\beta}\sin b = \frac{\Delta\sin\tilde{\beta} - y_e\cos\tilde{\beta}}{\sqrt{\Delta^2 + y_e^2}}$$

and therefore, with every step written out,

$$\dot{V} = U\, y_e\, \frac{\Delta \sin\tilde{\beta} - y_e \cos\tilde{\beta}}{\sqrt{\Delta^2 + y_e^2}} - \frac{U}{\gamma}\tilde{\beta}\,\dot{\hat{\beta}}$$

$$\dot{V} = \underbrace{\frac{U\,\Delta\, y_e \sin\tilde{\beta}}{\sqrt{\Delta^2 + y_e^2}} - \frac{U}{\gamma}\tilde{\beta}\,\dot{\hat{\beta}}}_{\text{indefinite — must be dealt with}} - \underbrace{\frac{U\, y_e^2 \cos\tilde{\beta}}{\sqrt{\Delta^2 + y_e^2}}}_{\text{negative for } |\tilde{\beta}|<90^\circ}$$

4-9-6. The adaptation law is forced, not chosen

- The second group is already what is wanted. The first group contains the unknown $\tilde{\beta}$ and has no sign, so the argument can only succeed if it is made to vanish. There is exactly one quantity still unspecified — $\dot{\hat{\beta}}$ — and one way to spend it.
- Replace $\sin\tilde{\beta}$ by $\tilde{\beta}$ for the moment, so that the two terms of the first group share the factor $\tilde{\beta}$:

$$\tilde{\beta}\left[\frac{U\Delta y_e}{\sqrt{\Delta^2 + y_e^2}} - \frac{U}{\gamma}\dot{\hat{\beta}}\right] = 0$$

- This must hold for **every** value of $\tilde{\beta}$, because $\tilde{\beta}$ is unknown — that is the entire premise of adaptive control. A condition that must hold for all $\tilde{\beta}$ forces the bracket itself to be zero, and the bracket contains only known quantities:

$$\frac{U}{\gamma}\dot{\hat{\beta}} = \frac{U\,\Delta\, y_e}{\sqrt{\Delta^2 + y_e^2}} \quad \Longrightarrow \quad \boxed{\dot{\hat{\beta}} = \gamma\,\frac{\Delta\, y_e}{\sqrt{\Delta^2 + y_e^2}}}$$

- **Note what the U did.** It cancelled. The adaptation law does not contain the speed, so it does not need a speed measurement — and this is why the Lyapunov weight had to carry U rather than being a bare $1/(2\gamma)$. Choosing the weight was the same act as choosing not to need a speedometer.
- Substituting back, what survives is

$$\dot{V} = \frac{U\Delta y_e\left(\sin\tilde{\beta} - \tilde{\beta}\right)}{\sqrt{\Delta^2 + y_e^2}} - \frac{U\, y_e^2 \cos\tilde{\beta}}{\sqrt{\Delta^2 + y_e^2}}$$

Term	Sign	Size
second	negative for $\lvert\tilde{\beta}\rvert < 90^\circ$	order y_e^2
first	either sign	order $\tilde{\beta}^3/6$, since $\sin\tilde{\beta} - \tilde{\beta} = -\tilde{\beta}^3/6 + O(\tilde{\beta}^5)$

- The first term is **third order** in the estimation error while the second is second order in y_e, so for small enough $\tilde{\beta}$ the negative term dominates and $\dot{V} < 0$. This is why the result is **uniform semiglobal exponential stability** and not a global one: the region of attraction is a ball whose size depends on how large an initial crab-angle error is admitted, and it does not extend to $\lvert\tilde{\beta}\rvert \geq 90^\circ$.
- The linear-in-$\tilde{\beta}$ step above is the only approximation in the derivation, and it is the reason for the word *semiglobal*. Fossen's own treatment reaches the same law and the same USGES conclusion.

Verification

Check	How	Result
Dimension	$\gamma \cdot [\mathrm{m}]$ must be rad/s	γ in rad/(m·s), as tabulated
Sign	$y_e > 0$ (starboard) must raise $\hat{\beta}$, since a starboard offset under an unmodelled current means β was underestimated	$\dot{\hat{\beta}} > 0$ for $y_e > 0$ — correct
Limit $y_e \to 0$	adaptation must stop when on the path	$\dot{\hat{\beta}} \to 0$
Limit $y_e \to \infty$	the rate must not run away	$\dot{\hat{\beta}} \to \gamma\Delta$, bounded — a built-in rate limit, as in §4-8-4
Numeric	measured in section G: does $\hat{\beta}$ converge to the true crab angle?	$\hat{\beta} = 15.91^\circ$ against a true $\beta_c = 15.88^\circ$ — an error of 0.04°
Numeric	settled cross-track error under current, section G	-0.004 m, against LOS's 2.253 m

- The estimate converging to the true crab angle to within 0.04° is the strongest available evidence that the law is right, since nothing in the law was ever told what the current was.

4-9-6a. Watching the estimate converge

- The same idealised integration as §4-8-5a, run with the ALOS law instead, and printed by the same script. Compare the two tables column by column: they are doing the same job with different bookkeeping.

t [s]	y_e [m]	$\hat{\beta}$ [deg]
0	0.000	0.00
10	1.602	3.01
25	1.203	9.32
50	0.323	14.28
100	0.013	15.69
200	0.000	**15.74**
400	0.000	**15.74**

What to notice	Why it happens
y_e grows to 1.60 m first	with $\hat{\beta} = 0$ the law *is* plain LOS, so the vessel drifts off exactly as in §4-7
$\hat{\beta}$ climbs **monotonically**, with no overshoot	the update $\dot{\hat{\beta}} = \gamma\Delta y_e / \sqrt{\Delta^2 + y_e^2}$ has the same sign as y_e, and y_e never changes sign here
$\hat{\beta} \to 15.74^\circ$	which is β_c exactly. **The law was never told the current speed or direction, and it recovered the drift angle to two decimals**
$y_e \to 0.000$	once the estimate is right, $\tilde{\beta} = 0$ and §4-9-4 reduces to plain LOS with no disturbance left

✎ **The one difference a beginner should take from the two tables**

ILOS's state ends at 7.516 — **seven and a half what?** Seconds, as §4-8-3 shows, and the number means nothing on its own.
ALOS's state ends at 15.74 **degrees**, and it is the angle the water is pushing the vessel through. Point a drift-angle sensor at the hull and it would read the same number.
Both vessels are on the path. Only one of them can say why.

4-9-7. The assumptions, and the honest picture of $V(t)$

Assumption	Used where	What fails if it does not hold
$\dot{\beta} = 0$ — constant current	$\dot{\tilde{\beta}} = -\dot{\hat{\beta}}$, §4-9-4	a time-varying current leaves a term $\frac{U}{\gamma}\tilde{\beta}\dot{\beta}$ in $\dot{V}$ with no sign; the estimate lags and V need not decrease
$\lvert\tilde{\beta}\rvert$ small	the linearisation of $\sin\tilde{\beta}$, §4-9-6	the cubic term can dominate; this is what makes the result semiglobal rather than global
$\lvert\tilde{\beta}\rvert < 90^\circ$	$\cos\tilde{\beta} > 0$	the leading negative term changes sign
$\psi = \psi_d$ — no autopilot lag	$\chi - \pi_p = \tilde{\beta} - \arctan(y_e/\Delta)$	**this one is violated in every real system**, including this week's, and the consequence is visible below
$U > 0$ and roughly constant	V's weight, and the differentiation of V	a decelerating vessel adds a $\dot{U}$ term to $\dot{V}$

⚠ $V(t)$ **measured on the real model is not monotone, and it should not be expected to be**

Section H computes V from the logged states of the full model. It **rises** from 5.80 to 12.94 over the first 17.5 s, then falls to 0.5588 — a net reduction by a factor of 23.2, but with only 62.2 % of samples decreasing.
The derivation above assumes $\psi = \psi_d$ instantly. The real vessel has a heading autopilot with finite bandwidth, so during the initial turn $\chi - \pi_p$ is set by where the bow actually is rather than by where the guidance law asked it to be, and the equation $\dot{y}_e = U\sin(\tilde{\beta} - \arctan(y_e/\Delta))$ — on which the whole $\dot{V} \le 0$ argument rests — does not yet describe the system.
This is not a defect in the derivation and not a bug in the model. It is the price of the cascade assumption, and quoting a monotone V from a simulation that does not produce one would be the actual error. The correct claim is the one the cascade literature makes: the guidance subsystem is USGES, the autopilot is exponentially stable, and the cascade of the two is stable — but V of the *guidance* subsystem alone is not a Lyapunov function for the *whole* system.

4-9-8. Choosing γ

- γ sets how fast the estimate moves. Too small and the vessel spends the leg still learning; too large and the estimate chases the transient rather than the current, which couples the estimator into the autopilot's own dynamics.
- Section H measures it, mean absolute cross-track error over the final 100 s:

γ [rad/(m·s)]	settled $\lvert y_e\rvert$ [m]	$\hat{\beta}$ [deg]	true crab [deg]
0.0005	0.9353	9.31	15.75
0.0010	0.4303	12.81	15.68
0.0020	0.0869	15.07	15.61
0.0050	**0.0059**	**15.91**	**15.88**
0.0200	0.1737	15.21	15.59

- Again an interior minimum, and this table shows *why* in a way the ILOS one cannot: **the estimate itself is observable.** At $\gamma = 0.0005$ the estimate is still 6.4° short of the true crab angle after 500 s — the adaptation simply has not finished. At $\gamma = 0.02$ it overshoots the other way and chases the corner transients instead of the current, so the estimate ends *further* from the truth than at $\gamma = 0.005$.
- Only at $\gamma = 0.005$ does the estimate land on the true crab angle, and that is also where the cross-track error is smallest. The two columns agreeing is the confirmation that the law converges for the reason the derivation says it does, rather than by accident.
- The default follows the measurement: $\gamma = 0.005$ **rad/(m·s)**.

4-9-9. ILOS against ALOS

	ILOS	ALOS
what the extra state is	y_{int}, an integral of scaled error — **no physical meaning**	$\hat{\beta}$, an estimate of the crab angle — **reads out in degrees**
unit of the state	s (heading form)	rad
tuning number	κ, in m/s	γ, in rad/(m·s)
removes the current by	accumulating until the offset it contributes matches the one the current caused	measuring the effect and cancelling the cause
stability	UGAS for the course form, by a quadratic V; cascade argument for the heading form	USGES — semiglobal, limited by $\lvert\tilde{\beta}\rvert < 90^\circ$
anti-windup	built into the denominator, §4-8-4	built into the bounded $\Delta y_e/\sqrt{\Delta^2 + y_e^2}$
measured settled y_e, section G	0.008 m	-0.004 m
diagnostic value	none — the state says nothing about the sea	the state **is** the crab angle, usable by other subsystems
reference implementation available	yes, MSS `ILOSpsi.m`	no, in the 2021 release used here

- On this mission they perform equivalently, and both are two orders of magnitude better than plain LOS. **The reason to prefer one is not accuracy.** ALOS produces a number that means something; ILOS has a proof that closes in one line and an implementation to check against.

4-10. Equation against MATLAB, side by side

- Every line of the guidance block is generated by `guidance_code(law)` inside `W04_1_build_guidance.m`, so the four laws share every line except the body. The tables below put each line beside the equation it implements.

The part all four laws share

Equation	MATLAB, as the block runs it
$\pi_p = \operatorname{atan2}(E_{k+1} - E_k,\ N_{k+1} - N_k)$	`pi_p = atan2(En - Ek, Nn - Nk);`
$x_e = \quad (N - N_k)\cos\pi_p + (E - E_k)\sin\pi_p$	`x_e = dN*cos(pi_p) + dE*sin(pi_p);`
$y_e = -(N - N_k)\sin\pi_p + (E - E_k)\cos\pi_p$	`y_e = -dN*sin(pi_p) + dE*cos(pi_p);`
$d_k = \lVert \mathbf{wp}_{k+1} - \mathbf{wp}_k \rVert$	`d = sqrt((Nn-Nk)^2 + (En-Ek)^2);`
$d_k - x_e < R$	`hit = (d - x_e) < R_switch;`
$\lVert \mathbf{p} - \mathbf{wp}_{k+1} \rVert < R$	`hit = sqrt((N-Nn)^2 + (E-En)^2) < R_switch;`
stop the index at $n-1$, §4-6	`if hit && k < n-1, k = k + 1; end`

The four bodies

Law	Equation	MATLAB
atan2	$\psi_d = \operatorname{atan2}(E_{k+1} - E,\ N_{k+1} - N)$	`psi_d = atan2(En - E, Nn - N);`
LOS	$\psi_d = \pi_p - \arctan(y_e/\Delta)$	`psi_d = pi_p - atan(y_e / Delta);`
ILOS	$\psi_d = \pi_p - \arctan(K_p y_e + K_i y_{int})$	`psi_d = pi_p - atan(Kp*y_e + Ki*y_int);`
	$\dot{y}_{int} = \dfrac{\Delta y_e}{\Delta^2 + (y_e + \kappa y_{int})^2}$	`y_int = y_int + h * Delta*y_e / (Delta^2 + (y_e + kappa*y_int)^2);`
ALOS	$\psi_d = \pi_p - \hat{\beta} - \arctan(y_e/\Delta)$	`psi_d = pi_p - b_hat - atan(y_e / Delta);`
	$\dot{\hat{\beta}} = \gamma \dfrac{\Delta y_e}{\sqrt{\Delta^2 + y_e^2}}$	`b_hat = b_hat + h * gamma * Delta * y_e / sqrt(Delta^2 + y_e^2);`

- The two update lines are **forward Euler**, `state = state + h * derivative`, exactly as MSS integrates its own integral state. The step h arrives in the parameter vector rather than being hard-coded, so changing the block's rate changes the integration consistently.

`persistent` inside a block is not `persistent` in a script

- Three quantities carry across time steps: the leg index k, the ILOS integral, and the ALOS estimate. All three are declared `persistent`, and this is the one place where the Simulink implementation behaves **better** than the MSS script it is derived from.

	MSS `ILOSpsi.m` called from a script	this week's MATLAB Function block
who owns the state	the **function**, one copy per MATLAB session	the **block**, one copy per block
running two laws at once	impossible — they would share `k` and `y_int`	four blocks, four independent copies
resetting between runs	`clear ILOSpsi` by hand, and forgetting it silently corrupts the next run	Simulink resets at the start of every simulation
the failure it causes	a second run starts on waypoint 4 with a full integrator	cannot occur

- This is the mechanism that makes the four-vessel comparison of Part 2 possible at all. Four rows of the same model run four laws simultaneously with the same waypoints, the same current and the same autopilot gains, and none of them can see another's state.

⚠ A block that carries memory needs a discrete sample time

A MATLAB Function block containing `persistent` cannot run under a continuous sample time: there is no well-defined "previous step" for the solver to carry the state across, and Simulink rejects the model. The guidance blocks are therefore declared discrete at rate h.

The rate is **not** a block parameter. A MATLAB Function block is a masked Stateflow object, so the rate is set through its Stateflow properties — `ChartUpdate = 'DISCRETE'` and `SampleTime = h` — which is what the fifth argument of `_tools/set_mlfcn.m` does. Setting `'SampleTime'` on the block itself raises `SubSystem block does not have a parameter named 'SampleTime'`.

Because h appears both as the block's rate and inside the Euler update, the two are guaranteed to agree. A block that ran at one rate and integrated with another would produce a law that is neither the continuous one nor the discrete one.

4-11. Beyond LOS

- LOS, ILOS and ALOS are the three laws this course implements. They are not the only ones, and a graduate reader should know what the alternatives buy and what they cost.

Approach	The idea	Buys	Costs	Reference
Curved paths (Hermite spline)	replace the straight leg by a monotone cubic spline; π_p becomes the spline's tangent angle at the closest point	continuous curvature, so no corner transient at all — the switching logic of §4-6 disappears	a closest-point search each step, and a curvature term in the error dynamics	Lekkas and Fossen (2014), *IEEE TCST* 22(6), 2287–2301
Vector field guidance	define a desired course at *every* point of the plane, not only along the path	provably no overshoot; extends naturally to 3-D and to orbits	the field must be designed, and the argument is no longer one arctan	Nelson et al. (2007), *IEEE Trans. Robotics* 23(3), 519–529
Model predictive path following	optimise the future track over a horizon subject to the actuator limits of Week 5	handles input saturation and obstacles *inside* the guidance layer	an optimisation every step; no closed-form law to verify by hand	Faulwasser and Findeisen (2016), *IEEE TAC* 61(4), 1026–1039
ALOS in 3-D	estimate both the sideslip and the angle of attack, for a vehicle that can dive	underwater path following with the same structure as §4-9	two adaptive states, and a coupled stability argument	Fossen and Aguiar (2024), *Ocean Engineering*

- The common thread is that **every one of them still consumes y_e and produces a course or heading command**. The interface built this week does not change; only the box between the two does. That is the practical value of keeping guidance and control in separate layers.

i What this week deliberately leaves out

Obstacle avoidance, collision regulations, and any form of replanning. Those change the *waypoint list*, which is a layer above guidance, not a change to the guidance law. Week 7 introduces the state machine that owns that layer.

Part 2 · Laboratory

- **One script per section.** There is no single runner. Each section below names the one file that produces its figure, and that file can be run on its own.
- Every script is in `lectures/W04_simulink/`. All of them drive the same model, `W04_guidance.slx`, and all of them read it back with the same `W04_read.m`.
- The shared files are `W04_vars.m` (every number), `W04_read.m` (unpacks the log), `W04_plot.m` (the standard track figure) and `W04_1_build_guidance.m` (builds the model from nothing).

A. Setting up

★ To produce every result in this section

```
cd lectures/W04_simulink
W04_0_setup
```

`W04_0_setup.m` is **the only file a student edits.** It copies every number out of `W04_vars.m` into the base workspace, so the model can be opened and run by hand.

Expected output:

```
W04 setup complete
  mission        5 waypoints, 4 legs, shortest 60.0 m
  guidance       Delta = 8 m, R_switch = 5 m, mode = 1 (along-track)
  feasibility    R_switch < shortest leg?  yes
  Delta / L      4.0 x hull length (rule of thumb: 2 to 5)
  ILOS           kappa = 0.3   ->  Ki = 0.0375
  ALOS           gamma = 0.005
  autopilot      Kp = 100, Kd = 74.9,  X_ff = 60 N
  current        0.00 m/s at 0 deg
  simulation     500 s at h = 0.02 s
```

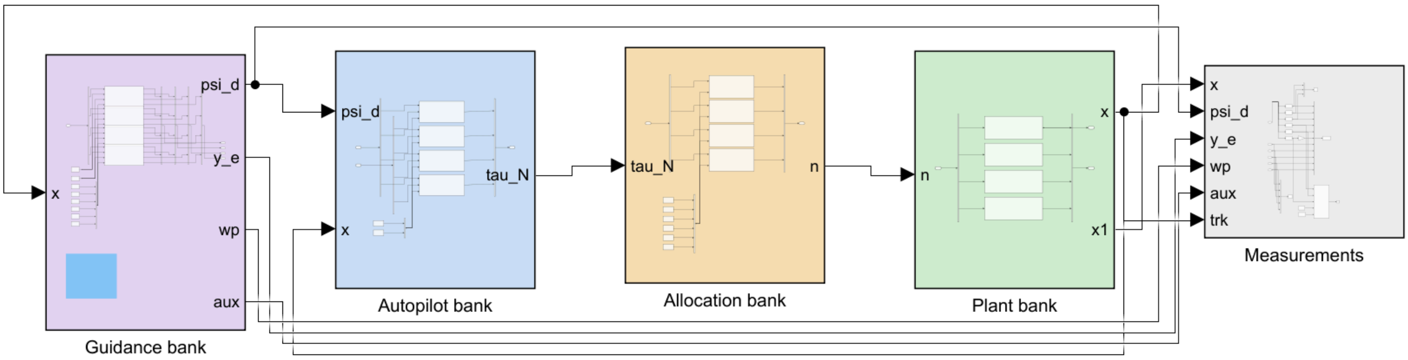
Line	What to check, and why
`shortest 60.0 m`	the mission is four legs: three sides of a square, then a diagonal, giving two 90° corners and one of 135°
`feasibility ... yes`	the §4-6 condition $R < \min_k d_k$. If this ever prints `NO`, stop: a waypoint will be skipped
`Delta / L 4.0`	the rule of thumb of §4-5, printed as a sanity check on a value that section E actually justifies
`Ki = 0.0375`	$\kappa/\Delta = 0.3/8$, the derived gain of §4-8-2 — not an independent number
`current 0.00 m/s`	sections C to F run in still water. G and H switch the current on

B. Building the model

★ To produce every figure in this section

```
cd lectures/W04_simulink
W04_1_build_guidance
```

This writes `W04_guidance.slx` and `img/W04_guidance.png` . Every other section calls it automatically if the model is missing, so it need only be run by hand after the model has been broken.



WEEK 4 - WAYPOINT FOLLOWING AND LOS GUIDANCE

Weeks 1 to 3 were told what heading to hold. Nobody said where the number came from. This week it comes from a list of waypoints, and turning that list into psi_d is what GUIDANCE means.

FOUR VESSELS RUN THE SAME MISSION AT THE SAME TIME

row 1 atan2 aim at the waypoint - and never reach the path
row 2 LOS aim Delta ahead on the path - and converge to it
row 3 ILOS LOS + an integral state - and beat the current
row 4 ALOS LOS + an estimated crab angle - and beat it knowingly

Same waypoints, same current, same autopilot gains, same hull. The only difference is the guidance block, so every difference in the tracks belongs to the guidance law. Set guid_show = 1..4 to plot one of them.

THE ONE SIGNAL THAT TRAVELS BACKWARDS

x, from the plant bank to the guidance bank. Guidance is feedback: it needs to know where the vessel IS before it can say where to point.

WHAT TO WATCH

1 WITH NO CURRENT, rows 2, 3 and 4 are nearly identical and row 1 is not. That is the difference between following a path and visiting points.

2 WITH A CURRENT, row 2 settles with a PERMANENT cross-track error of about Delta*tan(beta_c). Rows 3 and 4 remove it, by different means.

3 b_hat in row 4 converges to the crab angle itself. Read it off the scope and compare it with atan2(v, u).

Reading the figure

Element	Meaning
violet, **Guidance bank**	four rows, one per law. Takes the state x and produces ψ_d, y_e, the waypoint index, and the auxiliary state
blue, **Autopilot bank**	four identical copies of the Week 3 P–D law. Nothing here is retuned between rows
amber, **Allocation bank**	four copies of the square allocation of Appendix A1, turning τ_N into propeller speeds
green, **Plant bank**	four Otter hulls, identical, sharing one ocean current
grey, **Measurements**	the log. The first six columns are `[u v r N E psi]`, as in every week of this course
the one line going right to left	x, from the plant bank back to the guidance bank
blue annotation	what to watch, and why the four rows are laid out this way

- **The backward line is the whole point of the week.** Weeks 1 to 3 had a command that came from outside the model. Guidance is feedback: it cannot say where to point until it is told where the vessel is.
- The four rows share the waypoint list, the current, the autopilot gains and the hull. **The only difference between them is the guidance block**, so any difference in the tracks belongs to the guidance law and to nothing else.
- Set `guid_show = 1..4` in `W04_0_setup.m` to plot a single row instead of all four.

✎ Checking the model is still sound

```
check_overlaps('W04_guidance')
```

The pass mark is **0**. The function walks the top level and every subsystem, and reports any two signal lines that lie on top of each other. It ignores segments that share a source port, because a fan-out branch is meant to overlap.

C. Aiming at the waypoint

★ To produce every figure in this section

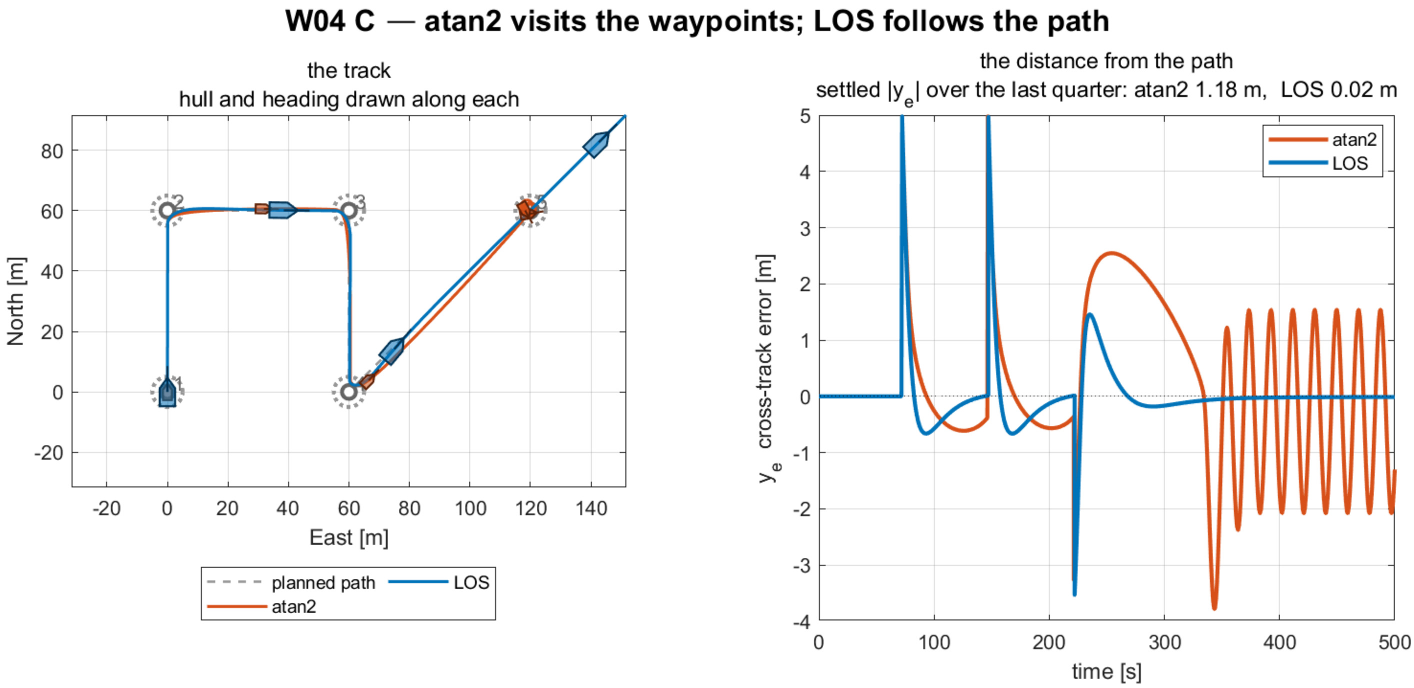
```
cd lectures/W04_simulink
W04_0_setup
W04_C_aim_at_the_waypoint
```

Produces `img/W04_result_atan2.png`.

Expected output:

```
W04 section C — psi_d = atan2(E_next - E, N_next - N)

  leg      atan2 |y_e|      LOS |y_e|      atan2 max         ratio
  ----------------------------------------------------------------------
  1             0.000          0.000          0.000      both zero
  2             0.547          0.060          0.611            9.1
  3             0.514          0.062          0.566            8.3
  4             1.184          0.014          2.077           86.7
```

Column	How it is measured
`atan2	y_e
`atan2 max`	the worst excursion anywhere on that leg
leg 1	both laws read exactly zero: the vessel starts on the path, already pointing along it, so there is nothing to correct and the ratio would be $0/0$



What the figure says

- **The meaning.** Two vessels, identical in every respect but the guidance law, run the same four-leg mission in still water. The left panel is where they went; the right is how far each was from the line it was supposed to be on.
- **The trend, with numbers.** On the three legs where the comparison is meaningful, `atan2` holds the path to 0.547, 0.514 and 1.184 m while LOS holds it to 0.060, 0.062 and 0.014 m — ratios of 9.1, 8.3 and 86.7. On the track panel the two vessels visit the same waypoints; only the blue one travels along the lines between them.
- **The principle.** Look at the shape of the orange curve after each corner. It rises to a peak and comes back down *slowly*, because the `atan2` law is not trying to return to the line — it is only trying to reduce the distance to the next waypoint, and drifting sideways barely changes that distance. The blue curve is pulled back within one look-ahead length because the arctan term grows with exactly the quantity that is wrong.
- **The difference between the algorithms.** They differ in one line of code. `atan2` measures to a point, LOS measures to a line. Everything in the right panel follows from that.
- **The part that is easy to miss.** After about 340 s the orange curve breaks into a **sustained oscillation** of roughly ± 2 m that never decays. That is not noise and not a plotting artefact: the vessel has passed the final waypoint at about 332 s, so the point it is chasing is now **behind it**. The `atan2` command flips by 180°, the vessel turns around, overshoots the waypoint again, and repeats — a limit cycle produced entirely by the guidance law. It is also why the leg-4 entry of 1.184 m in the table above should be read as an oscillation amplitude rather than as a tracking error.
- **What LOS does instead** at the same moment is nothing at all: the aim point is Δ ahead on the *extension* of the last leg, so it stays ahead of the vessel for ever and the blue curve simply stays flat. §4-6 notes that neither behaviour is a mission-termination policy; this figure is what the absence of one looks like.

D. The line-of-sight law

★ To produce every figure in this section

```
cd lectures/W04_simulink
W04_0_setup
W04_D_line_of_sight
```

Produces `img/W04_result_los.png`.

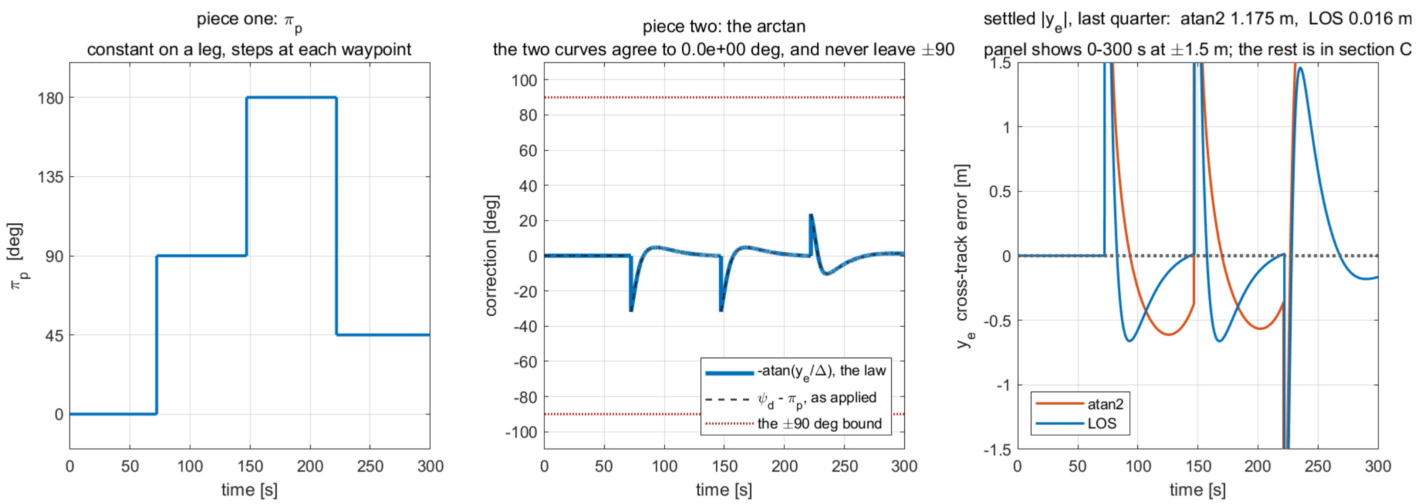
Expected output:

```
W04 section D — psi_d = pi_p - atan(y_e / Delta)

  Delta = 8 m,  path-tangential angle on leg 1 = 0.0 deg

  recomputed psi_d against the logged one, leg 1:  0.000e+00 deg
  (the same arithmetic, so this is a check on the LOG, not the law)

  law         settled |y_e|        max |y_e|        RMS |y_e|
  ------------------------------------------------------------
  atan2              1.1753           4.9800           1.4489
  LOS                0.0159           4.9799           0.5670
```

Quantity	Interval it is measured over
`settled	y_e
`max	y_e
`RMS	y_e

- The first number is a check on the **logging**, not on the law: ψ_d is recomputed here from the logged y_e using the equation of §4-4, and compared with the logged ψ_d. Agreement to $0.000\mathrm{e}{+}00$ confirms that the signal named `y_e` in the log really is the one the block used.

W04 D — one arctan turns waypoint-chasing into path following



What the figure says

- **The meaning.** The three panels take the law $\psi_d = \pi_p - \arctan(y_e/\Delta)$ apart into its two terms and then show what the pair achieves.

- **Panel 1, the trend.** π_p is a **staircase**: 0°, then 90°, then 180°, then 45°, changing only at the three switching instants near 72, 147 and 222 s and constant in between. This is the geometry of the mission and it contains no feedback whatever.
- **Panel 2, the principle.** Two curves are drawn: the correction the law *prescribes*, $-\arctan(y_e/\Delta)$, and the correction actually applied, $\psi_d - \pi_p$. **They lie exactly on top of each other** — agreement to $0.0\mathrm{e}{+}00$ degrees — which is the law verified graphically rather than argued. The correction is zero on the straight parts, dips to about -33° at each of the first two corners and reaches $+23^\circ$ at the 135° corner, and **never approaches the $\pm 90^\circ$ bound** drawn in red. That bound is the guarantee of §4-4: the vessel can be told to head straight at the path but never away from it.
- **Panel 3, the difference between the algorithms.** Zoomed to ± 1.5 m, the same corner transients are visible for both laws — they start from the same place and take a comparable excursion — but LOS returns to the line and stays there while `atan2` does not. Settled over the last quarter of the run, that is 0.016 m against 1.175 m, a factor of **seventy-four**.
- **Why the panels stop at 300 s** while the settled numbers are measured over 375–500 s: the last quarter of the run is where `atan2` is in the limit cycle of section C, which would compress everything else into a band. The comparison of settled values belongs here; the picture of the limit cycle belongs to section C.

E. The look-ahead distance

★ To produce every figure in this section

```
cd lectures/W04_simulink
W04_0_setup
W04_E_lookahead_distance
```

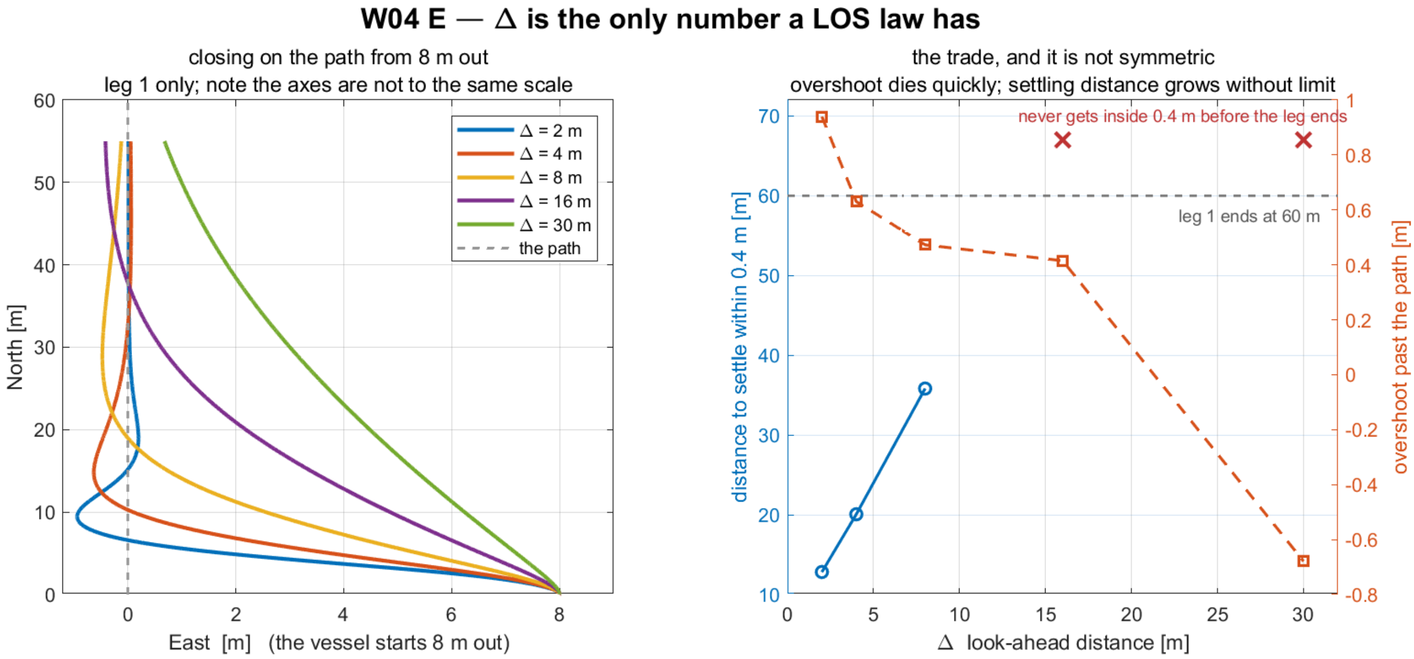
Produces `img/W04_result_lookahead.png` . Five runs, one per value of Δ.

- The vessel is started 8 **m off the path** in every run, so each has the same disturbance to recover from. A run that starts on the path measures nothing about how hard the law pulls.

Expected output:

```
   Delta  Delta/L     settle [m]  overshoot [m]  settled |y_e|  max |psi_d'|
-----------------------------------------------------------------------------------
       2      1.0          12.82          0.939         0.0134          14.64
       4      2.0          20.06          0.630         0.0559           5.76
       8      4.0          35.82          0.473         0.2321           2.23
      16      8.0    not on leg 1          0.415         0.2796           0.80
      30     15.0    not on leg 1         -0.679         1.2871           0.27
```

Column	Definition
`settle [m]`	how far **North** the vessel had travelled when $\lvert y_e \rvert$ last left the 0.4 m band. A distance rather than a time, because the speed is identical in every run and distance is what a chart shows
`not on leg 1`	the vessel never entered the band before the 60 m leg ended. There is no settling distance, and the table says so rather than quoting the leg length as though it were one
`overshoot [m]`	the furthest the vessel went **past** the path. A negative value means it never crossed
`max	psi_d'



What the figure says

- **The meaning.** Left: five approaches to the same path from the same 8 m offset, one per Δ. Right: the two costs plotted against Δ, on twin axes.
- **The trend, with numbers.** On the left, $\Delta = 2$ m (blue) turns almost perpendicular to the path, reaches it by $N \approx 13$ m and **crosses it**, swinging 0.94 m to the far side before recovering. $\Delta = 30$ m (green) leans so gently that at the end of the 60 m leg it is still 1.29 m out and has never crossed. Between them the curves fan out monotonically.
- **The principle.** All five vessels obey the identical law and differ only in one length. The arctan converts the *ratio* y_e/Δ into an angle, so a small Δ saturates that ratio at once — the command sits near 90° for most of the approach, which is why `max |psi_d'|` is 14.64 deg/s at $\Delta = 2$ and 0.27 deg/s at $\Delta = 30$, a factor of fifty.
- **The difference between the situations.** The right panel shows the trade is **not symmetric**. Overshoot falls quickly and then flattens: going from $\Delta = 2$ to 4 buys 0.31 m of it, but going from 8 to 16 buys only 0.06 m. Settling distance, by contrast, grows without limit — 12.82, 20.06, 35.82 m and then off the end of the leg entirely. **Past about $\Delta = 8$ m there is nothing left to buy and a great deal still to pay.**
- **The two red crosses** are drawn above the dashed line marking the end of leg 1, which is where "no settling distance" honestly belongs: those two runs did not fail to be measured, they failed to settle.
- **Note the axes on the left panel are not to the same scale.** The leg is 60 m long and the whole story happens within 9 m of East. Drawn to a true aspect ratio, all five tracks collapse onto the path line and the figure shows nothing.

F. Waypoint switching

★ To produce every figure in this section

```
cd lectures/W04_simulink
W04_0_setup
W04_F_waypoint_switching
```

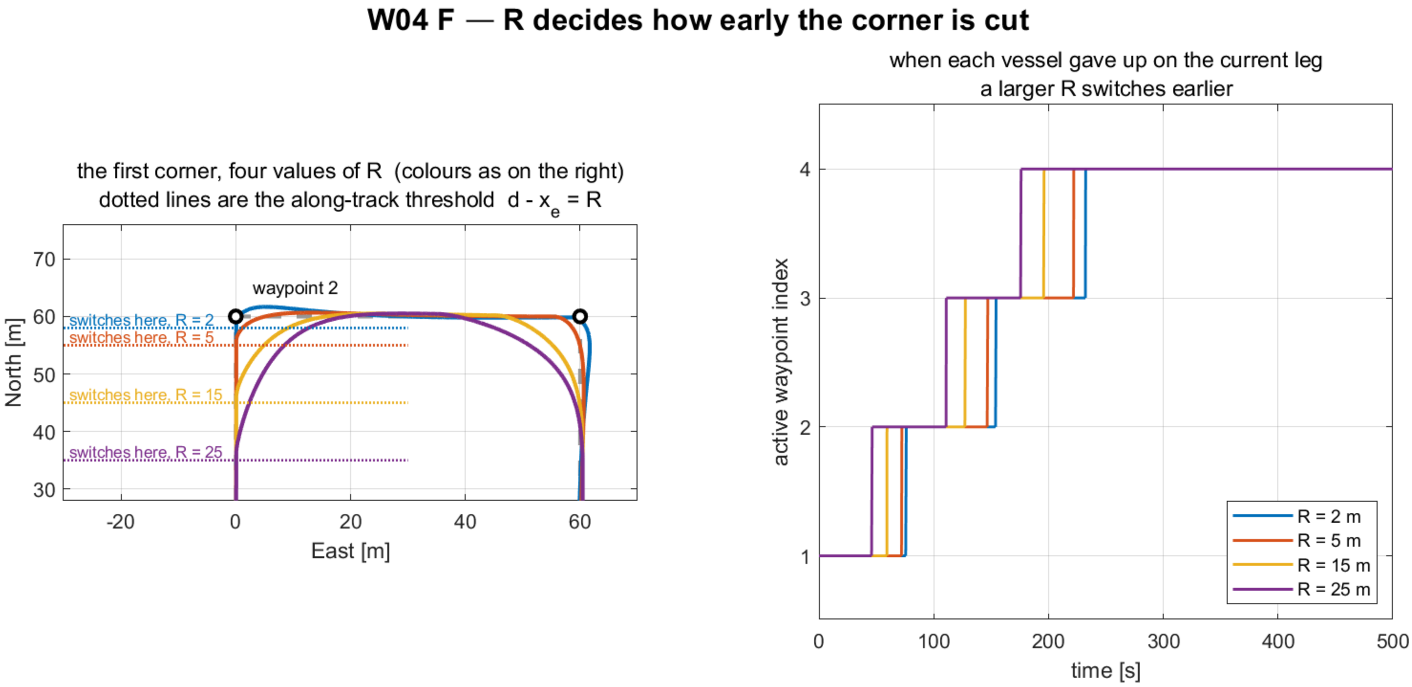
Produces `img/W04_result_switching.png`. Four runs, one per value of R.

Expected output:

```
A vessel 4.5 m short of waypoint 2 along the leg, 10.0 m to the side:

  along-track,  d - x_e = 4.50 < R = 5   ->  SWITCH
  circle,      |P - wp| = 10.97 < R = 5   ->  hold

   R [m] corner cut [m] switch times [s]    settled |y_e|
------------------------------------------------------------
       2           0.44     [76 154 232]            0.004
       5           2.26     [72 147 222]            0.016
      15           7.02     [59 128 196]            0.019
      25           9.62     [46 111 176]            0.018
```

- The first block is §4-6 as arithmetic. One position, two tests, two different answers: the along-track remainder is 4.50 m and passes; the true distance is 10.97 m and fails by more than a factor of two.

Column	Definition
`corner cut [m]`	the **closest** the vessel came to waypoint 2. A large R turns early and misses the corner by more
`switch times [s]`	when each of the three switches occurred
`settled	y_e



What the figure says

- **The meaning.** Left: the first 90° corner, with the four tracks and the four **switching thresholds** drawn as horizontal dotted lines. Right: the active waypoint index against time, for the same four runs.
- **The thresholds are lines, not circles.** Leg 1 runs due North along $E = 0$, so the along-track test $d_k - x_e < R$ is the half-plane $N > 60 - R$, whose boundary is horizontal. This is the distinction §4-6 makes, drawn on the data: had circles been drawn here, the figure would contradict the page that precedes it.
- **The trend, with numbers.** Each track leaves the leg exactly where its own dotted line crosses it, and the corner cut follows: 0.44 m at $R = 2$, then 2.26, 7.02 and 9.62 m. The $R = 25$ track (purple) begins turning at $N = 35$ m, fully 25 m before the waypoint, and passes almost 10 m from it.
- **The principle.** R does not tune accuracy. It decides **how early the vessel gives up on the current leg**, and the corner cut is the direct consequence.

- **The right panel** makes the same statement in time: all four staircases have the same three steps, shifted earlier as R grows — $[76\ 154\ 232]$ s at $R = 2$ against $[46\ 111\ 176]$ s at $R = 25$. Cutting the corners saves 56 s over the mission.
- **The difference between the situations.** The last column of the table barely moves: 0.004 to 0.019 m. R **costs almost nothing in straight-line accuracy** — it is paid for entirely at the corners. A survey that needs each line held to its end wants a small R; a transit that only needs to pass near the waypoints wants a large one. Neither is wrong.

G. The current, and the two laws that beat it

★ To produce every figure in this section

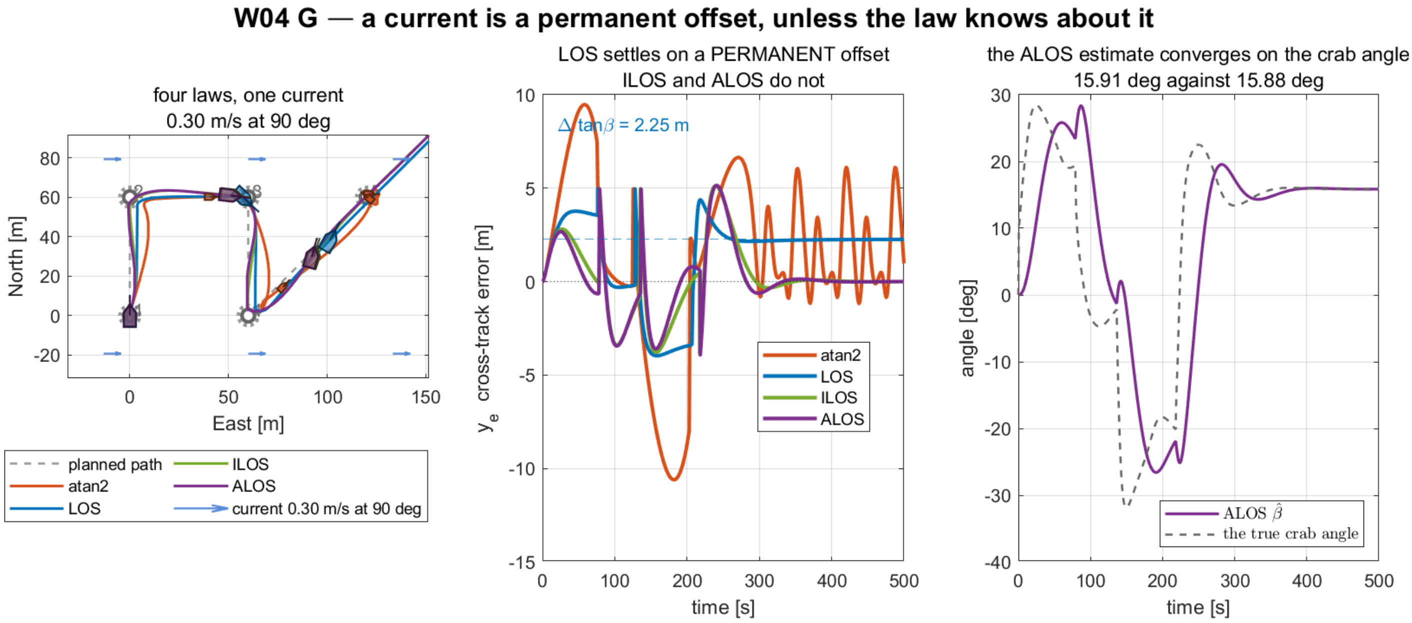
```
cd lectures/W04_simulink
W04_0_setup
W04_G_current_and_integral
```

Produces `img/W04_result_current.png`. One four-vessel run, a six-point sweep of the current speed, and finally the two state-filling tables printed in §4-8-5a and §4-9-6a.

Expected output:

```
W04 section G — a current of 0.30 m/s at 90 deg

  law         settled y_e     crab [deg]      aux state
  --------------------------------------------------------
  atan2             1.910          -4.60          0.000
  LOS               2.253          15.74          0.000
  ILOS              0.008          15.77          7.519
  ALOS             -0.004          15.88          0.278

    Delta tan(beta) = 8.0 x tan(15.74 deg) = 2.2545 m
    measured                                = 2.2533 m
    difference                              = 0.0012 m

       V_c      crab [deg]        LOS y_e       predicted       ILOS y_e
  ----------------------------------------------------------------------
      0.00          -0.10          -0.014          -0.013          -0.019
      0.10           5.10           0.713           0.714          -0.007
      0.20          10.37           1.463           1.464          -0.001
      0.30          15.74           2.253           2.254           0.008
      0.40          21.26           3.111           3.112           0.035
      0.50          27.38           4.145           4.143           0.054
```

Row	What it confirms
`LOS 2.253`	the §4-7 prediction $y_e^{ss} = \Delta \tan \beta_c$, to 0.0012 m
`ILOS 0.008` with `aux 7.519`	the §4-8-5 equilibrium: $y_{int}^{eq} = \Delta \tan \beta_c / \kappa = 7.5158$ s predicted, 7.519 s measured
`ALOS -0.004` with `aux 0.278`	0.278 rad is 15.91°, against a true crab angle of 15.88°
the sweep	the prediction holds at every current speed, to at most 0.002 m



What the figure says

- **The meaning.** All four laws, one mission, one current of 0.30 m/s from the East. Left: the tracks, with the current drawn as arrows. Middle: the cross-track error of each. Right: what the ALOS vessel believes the crab angle to be, against what it actually is.
- **The trend, with numbers.** In the middle panel the blue LOS curve settles onto the **dashed grey line**, which is not fitted to the data — it is $\Delta \tan \beta_c = 2.25$ m, drawn from §4-7 before the run. Green (ILOS) and purple (ALOS) converge to zero instead: 0.008 m and -0.004 m. Orange (`atan2`) is in its limit cycle again, now with the current widening it to about ± 6 m.
- **The principle.** The blue curve is the most important line in the week. **LOS has not failed and is not mistuned**: its heading error is zero, its autopilot is doing exactly what it was told, and it still sits 2.25 m off the path, because the law commands a heading while the vessel travels along a course. The offset is what the law asks for.
- **The difference between the algorithms.** ILOS and ALOS remove the same offset by opposite routes. ILOS accumulates an integral state until it happens to cancel the error, and the state settles at 7.519 — **a number with no physical meaning**, in units of seconds. ALOS estimates the crab angle itself, and its state settles at 15.91° against a true 15.88° — **a number that can be read off and checked against $\operatorname{atan2}(v, u)$.** Both end within 0.01 m of the path; only one can say why.
- **The right panel repays close reading.** The dashed line — the true crab angle — swings between $+28^\circ$ and -32° during the first 300 s. That is not disturbance: the current comes from a fixed direction, so as the vessel turns through each corner the current strikes the hull at a different angle and the crab angle genuinely changes with it. The estimate (purple) follows with a visible **lag**, which is what a first-order adaptation law does. Only on the final leg, where the heading is constant, do the two settle together at about 15.9°.
- **The difference between the situations.** The estimate is good when the assumption of §4-9-7 holds — a constant β — and lags whenever it does not. The sweep table shows the other side of the same coin: the ILOS residual grows slowly with current speed, from -0.019 m at rest to 0.054 m at 0.5 m/s, because a larger current means a larger integral state and a longer time to reach it within a fixed run.

H. The gains, and the Lyapunov function

★ To produce every figure in this section

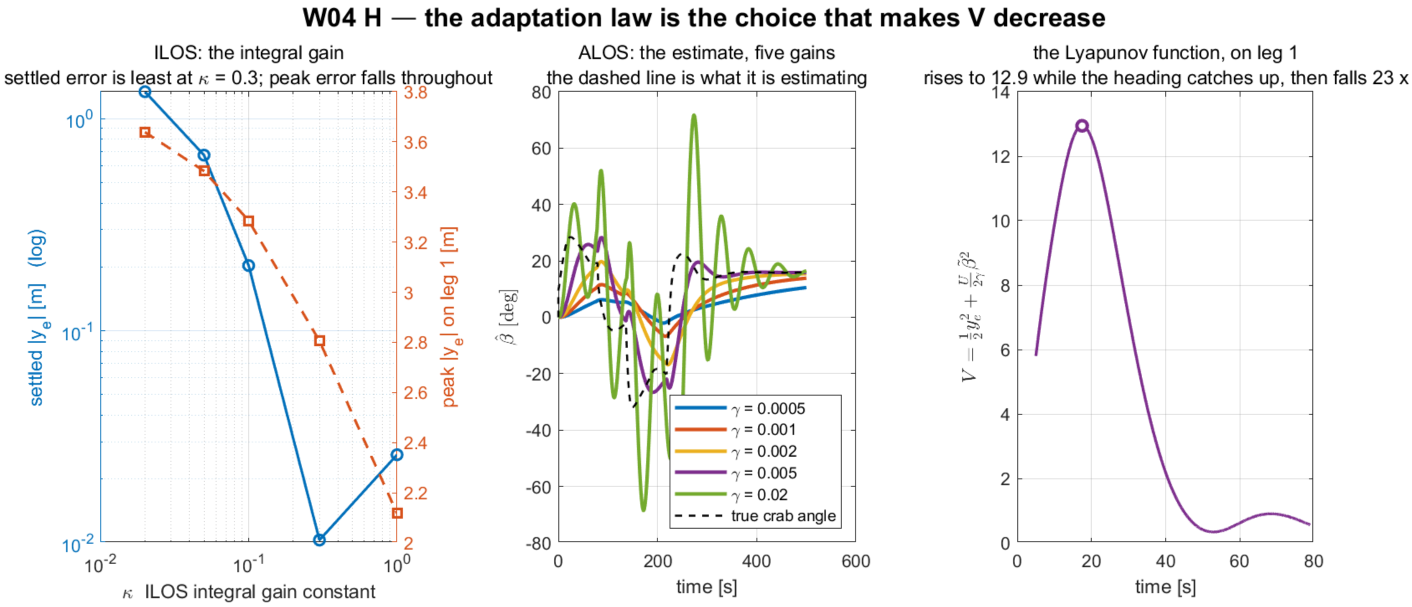
```
cd lectures/W04_simulink
W04_0_setup
W04_H_adaptive_and_stability
```

Produces `img/W04_result_stability.png`. Ten runs — five values of κ, five of γ — plus the Lyapunov computation on leg 1.

Expected output:

```
  1) ILOS: kappa, with Ki = kappa / Delta

       kappa          Ki     settled |y_e|       peak |y_e|     settling [s]
    -----------------------------------------------------------------------
        0.02     0.0025          1.3425          3.6388              NaN
        0.05     0.0063          0.6742          3.4835              NaN
        0.10     0.0125          0.2033          3.2855              NaN
        0.30     0.0375          0.0103          2.8057             66.9
        1.00     0.1250          0.0260          2.1178             73.2

  2) ALOS: gamma

         gamma    settled |y_e|      b_hat [deg]  true crab [deg]     settling [s]
    ---------------------------------------------------------------------------------
        0.0005          0.9353            9.31            15.75              NaN
        0.0010          0.4303           12.81            15.68              NaN
        0.0020          0.0869           15.07            15.61              NaN
        0.0050          0.0059           15.91            15.88              NaN
        0.0200          0.1737           15.21            15.59              NaN

  3) the Lyapunov function on leg 1

    V at the start of the leg  5.8005
    V at its peak              12.9433  at t = 17.5 s
    V at the end of the leg    0.5588
    fall from the peak         23.2 x
    samples with dV <= 0       62.2 %
```

- A `NaN` in the settling column means the run never met the settling criterion inside the leg. For the small gains that is because the state was **still moving**, not because anything diverged.



What the figure says

- **The meaning.** Left: the ILOS gain swept over five values, on log axes. Middle: the ALOS estimate for five values of γ, against the truth. Right: the Lyapunov function of §4-9-5, evaluated on the real vessel over leg 1.
- **Panel 1, the trend.** The settled error (blue, log scale) falls from 1.34 m at $\kappa = 0.02$ to 0.0103 m at $\kappa = 0.3$ and then **rises again** to 0.0260 m at $\kappa = 1.0$. The minimum is real and is the reason this course uses $\kappa = 0.3$. A linear axis would have hidden it: the last three points would all have sat on the zero line.
- **Panel 1, the principle.** The peak error (orange, right axis) falls monotonically throughout, from 3.64 m to 2.12 m. **The two curves disagree about which gain is best**, and that is the trade: a larger κ pulls harder, so it reduces the worst excursion while degrading the settled value. Choosing κ means choosing which of those two the mission cares about.
- **Panel 2, the difference between the algorithms' settings.** The dashed black line is the true crab angle; the five coloured curves are what each γ believes it to be. $\gamma = 0.0005$ (blue) crawls, reaching only 9.31° after 500 s — the adaptation is simply too slow to finish. $\gamma = 0.02$ (green) does the opposite, swinging between $+70^\circ$ and -70°: it is tracking the *corner transients* rather than the current. **Those excursions also break the assumption the derivation rests on**, since §4-9-7 requires $|\tilde{\beta}| < 90^\circ$ and this is within sight of it. $\gamma = 0.005$ (purple) follows the truth with a modest lag and settles on it.
- **Panel 3, the honest result.** V starts at 5.80, **rises to 12.94 at $t = 17.5$ s**, and then falls to 0.5588 — a reduction of $23.2\times$ from the peak, with only 62.2 % of samples decreasing.
- **The principle behind the rise.** §4-9-6 proved $\dot{V} < 0$ *on the assumption that $\psi = \psi_d$ exactly*. There is a Week 3 autopilot in between, with its own settling time. During the first seconds of the leg the heading is still catching up, so $\chi - \pi_p$ is set by where the bow actually is rather than by where the guidance asked it to be, and the error dynamics on which the proof rests do not yet describe the vessel. Once the heading has converged, V falls monotonically for the rest of the leg.
- **Why this is reported rather than hidden.** A monotone V could have been produced by plotting the kinematic subsystem in isolation, and it would have been a picture of an assumption rather than of a vessel. The claim the reference actually supports is that the guidance subsystem is USGES and that the cascade of it with a stable autopilot is stable — **not** that V of the outer loop alone decreases at every instant of a real run.

Summary

Week Summary

Step	What was done	How it was verified
§4-2, §4-3	derived π_p and the pair (x_e, y_e) from one rotation	compared against MSS `crosstrackWpt.m` at eight test points; largest disagreement 0
§4-4	derived the LOS law from the aim-point geometry	five limits checked numerically; compared against MSS `LOSchi.m`; recomputed from the log in section D, agreement $0.0\text{e}{+}00$ deg
§4-5	fixed $\Delta = 8$ m	five-point sweep, section E: settling distance and overshoot move in opposite directions and 8 m is where the second stops improving
§4-6	established the two switching criteria and $R < \min_k d_k$	section F evaluates both tests on one position — 4.50 m passes, 10.97 m fails; four-point sweep of R
§4-7	derived $\dot{y}_e = U \sin(\chi - \pi_p)$ and $y_e^{ss} = \Delta \tan \beta_c$	six-point current sweep, section G; worst disagreement 0.002 m on a 4.1 m offset
§4-8	derived ILOS, its units, its built-in anti-windup and its equilibrium	$y_{int}^{eq} = \Delta \tan \beta_c / \kappa$ predicted 7.5158 s, measured 7.519 s; closed form of $\dot{y}_e$ checked to 6.7×10^{-16} over 10^4 states
§4-8-6	showed the course normalisation admits a constant-weight Lyapunov function and the heading one does not	$\dot{V}$ matches $-U \cos \beta_c y_e^2 / D$ to 1.1×10^{-14}; the heading form leaves a residual of rms 2.94
§4-9	derived ALOS and showed the adaptation law is forced	$\hat{\beta} = 15.91^\circ$ against a true 15.88°, with nothing in the law told what the current was
§4-10	put every equation beside the line of MATLAB that implements it	the code shown is generated by `guidance_code(law)`, so it cannot drift from the model
Part 2	eight scripts, six result figures	every number quoted in this document appears in the console output of the script named above it

Progress Check

- [ ] `W04_0_setup` prints `feasibility ... yes` and `Ki = 0.0375`.
- [ ] `check_overlaps('W04_guidance')` returns **0**.
- [ ] Section C reproduces the leg-2 ratio of 9.1 between `atan2` and LOS.
- [ ] Section D reports the recomputed ψ_d agreeing with the logged one to $0.000\text{e}{+}00$ deg.
- [ ] Section E shows settling distance rising and overshoot falling as Δ grows, with two values that never settle.
- [ ] Section F prints $4.50 < 5$ passing the along-track test while 10.97 fails the circle test.
- [ ] Section G reproduces $\Delta \tan \beta_c = 2.2545$ m against a measured 2.2533 m.
- [ ] Section H shows the ILOS settled error with a minimum at $\kappa = 0.3$ and the ALOS estimate landing on the true crab angle at $\gamma = 0.005$.
- [ ] The learner can state, without looking, which of x_e and y_e the switching logic uses and why.
- [ ] The learner can explain why $V(t)$ in section H is not monotone.

Assignment 4

① Requirements

1. **Add a fifth row to the guidance bank** implementing the *course-angle* ILOS of MSS `ILOSchi.m`, that is, with the normalisation $\dot{y}_{int} = U y_e / \sqrt{\Delta^2 + (y_e + \kappa y_{int})^2}$ and the command applied to χ rather than ψ. Note that κ is now **dimensionless**, so the value 0.3 from this week must not be carried across unchanged; state the correct scaling and justify it.
2. **Sweep the current direction** $\beta_c \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ\}$ at a fixed $V_c = 0.3$ m/s, for LOS, ILOS and ALOS. Report the settled $|y_e|$ for each of the fifteen combinations in one table.
3. **Change the mission** so that one leg is shorter than R_{switch} and show what `wp_switch.m` does. Do not remove the check; demonstrate it firing.

② Verification (required — an unverified result scores zero)

Each of the following must be reported as a number produced by a script, not as an assertion:

Claim	Required evidence
the fifth row implements the course law	its ψ_d recomputed from the logged y_e and y_{int}, agreeing to better than 10^{-9} deg
the rescaled κ is right	a dimensional argument in the report **and** an equilibrium check: κy_{int}^{eq} against $\Delta \tan \beta_c$
the current sweep	for the LOS rows, each measured offset against $\Delta \tan \beta_c$ computed from the *measured* β_c of that run
the short-leg case	the error text, and an explanation of which inequality was violated
the model is still readable	`check_overlaps` returning 0

③ Analysis

Answer in at most two pages:

- At $\beta_c = 0^\circ$ and 180° the current is along the path. Predict the settled offset for all three laws **before** running, then compare. Explain any discrepancy.
- ILOS and ALOS reach almost identical settled errors in section G. Give two circumstances in which they would **not**, and state which law is preferable in each.
- Section H shows V rising for the first 17.5 s. Propose a change to the *model* — not to the guidance law — that would reduce that rise, and predict what it would cost.

Item	Marks
① requirements, all three implemented and running	30
② verification, every row evidenced	40
③ analysis, with predictions made before measurement	25
presentation: figures readable, numbers traceable to a script	5

Troubleshooting

Only problems actually encountered while building this week.

Symptom	Cause	Fix
`Invalid setting for parameter 'Value' ... 'mp'` when building	Constant blocks hold variable **names**, which Simulink validates at creation, and the base workspace was empty	`ensure_base_vars(W04_vars)` at the top of the builder
`SubSystem block does not have a parameter named 'SampleTime'`	a MATLAB Function block is a masked Stateflow object; the rate is a chart property	set `ChartUpdate = 'DISCRETE'` and `SampleTime` on the chart — the fifth argument of `set_mlfcn`
`Persistent variables are not supported with continuous sample time`	the guidance block carries k, y_{int} and $\hat{\beta}$ across steps	give it the discrete rate h, as above
Measurements port width 48 against 12	the plant bank exposed only the stacked state	added the separate `x1` output
`check_overlaps` reported 572 overlapping segments	32 separate constant-to-block lines crossing the diagram	collect the tuning into the vectors `par`, `gains` and `apar`, and route them through one lane each
`SampleTime 'h'` suddenly invalid, with no edit to the model	an ad-hoc diagnostic in the base workspace had overwritten `h` with a port-handle struct	never reuse a model variable's name in a scratch command; re-run `W04_0_setup`
A section's figure looks identical to the previous section's	`W04_plot` was called for both	it is a defect, not a style choice — section D now plots the decomposition of the law instead

References

Source	Where
Fossen, T. I. (2021). *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley	§12.3 path following; §10.3 the Otter
Børhaug, E., Pavlov, A. and Pettersen, K. Y. (2008). Integral LOS control for path following of underactuated marine surface vessels in the presence of constant ocean currents	*Proc. 47th IEEE CDC*, 4984–4991. The ILOS law of §4-8
Lekkas, A. M. and Fossen, T. I. (2014). Integral LOS path following for curved paths based on a monotone cubic Hermite spline parametrization	*IEEE TCST* 22(6), 2287–2301. The course-angle ILOS and the curved-path extension
Nelson, D. R. et al. (2007). Vector field path following for miniature air vehicles	*IEEE Trans. Robotics* 23(3), 519–529. §4-11
Faulwasser, T. and Findeisen, R. (2016). Nonlinear model predictive control for constrained output path following	*IEEE TAC* 61(4), 1026–1039. §4-11
MSS `ILOSpsi.m`, `ILOSchi.m`, `LOSchi.m`, `crosstrackWpt.m`	the vendored 2021 release, located by `_tools/mss_path.m`

⚠ **One reference this week does not have**

The vendored MSS is the 2021 release and contains no `ALOSpsi.m`; adaptive LOS entered MSS in 2023. The ALOS derivation of §4-9 was therefore carried out from first principles in this document and verified numerically — by the convergence of $\hat{\beta}$ to the true crab angle — rather than checked against a reference implementation. Where it differs in scaling from Fossen (2023), the derivation is what this course stands behind, and §4-9 says so at the top.

Next Week

Week 5 — Control Allocation. This week gave all four vessels the same square allocation from Appendix A1 and never questioned it. Week 5 asks what happens when there are more actuators than degrees of freedom, when a thruster saturates, and when one fails outright — and turns $\boldsymbol{\tau} = \mathbf{B}\mathbf{f}$ from a matrix inverse into an optimisation.